

Statistical Machine Learning (BE4M33SSU)

Predictor evaluation

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Prediction problem formulation

- ◆ We want to find a predictor (strategy, hypothesis, classifier) $h: \mathcal{X} \rightarrow \mathcal{Y}$
- ◆ Single prediction evaluated by loss function $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$
- ◆ The performance of h is evaluated by expected risk

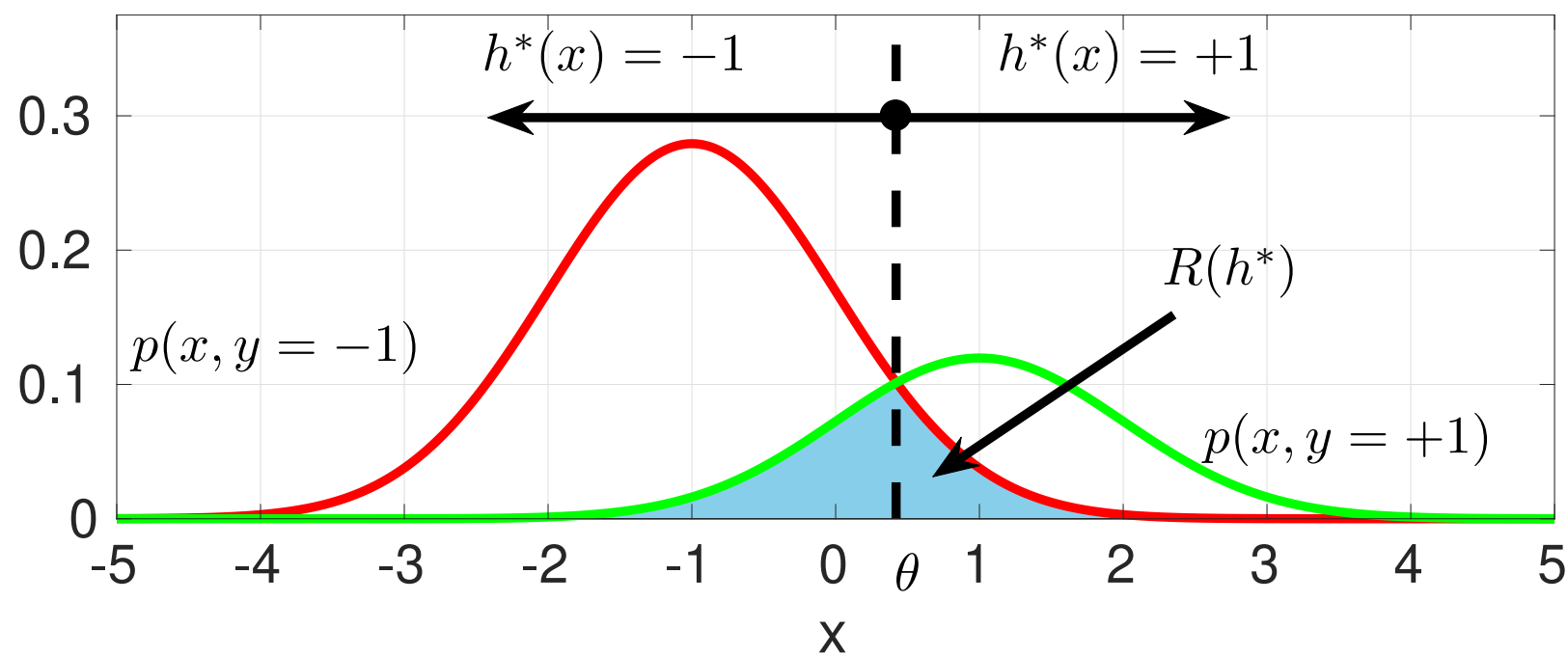
$$R(h) = \int \sum_{y \in \mathcal{Y}} \ell(y, h(x)) p(x, y) \, dx = \mathbb{E}_{(x, y) \sim p} [\ell(y, h(x))]$$

The expected risk $R(h)$ represents the average error of h when evaluated over i.i.d. sample $(x^1, y^1), \dots, (x^n, y^n)$ with $n \rightarrow \infty$:

$$\frac{\ell(y^1, h(x^1)) + \ell(y^2, h(x^2)) + \dots + \ell(y^n, h(x^n))}{n} \xrightarrow{p} R(h)$$

- ◆ The Bayes (optimal) predictor: $h^* \in \min_{h \in \mathcal{Y}^{\mathcal{X}}} R(h)$

Example of a prediction problem



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◆ The statistical model is known:

- $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{+1, -1\}$, $\ell(y, y') = \begin{cases} 0 & \text{if } y = y' \\ 1 & \text{if } y \neq y' \end{cases}$
- $p(x, y) = p(y) \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu_y)^2}$, $y \in \mathcal{Y}$.

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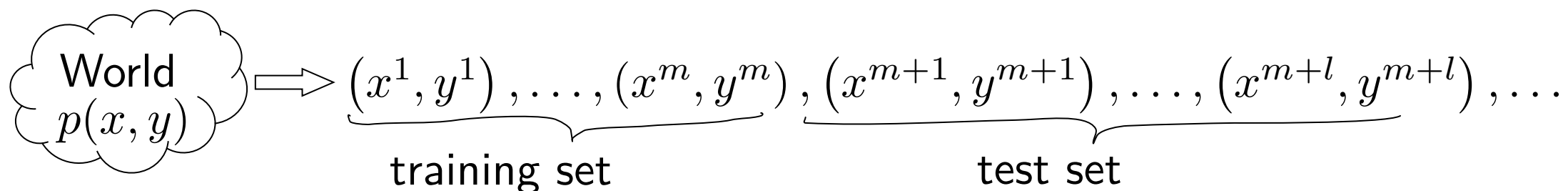
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◆ The value of the true risk:

$$R(h^*) = \int_{-\infty}^{\theta} p(x, +1) dx + \int_{\theta}^{\infty} p(x, -1) dx$$

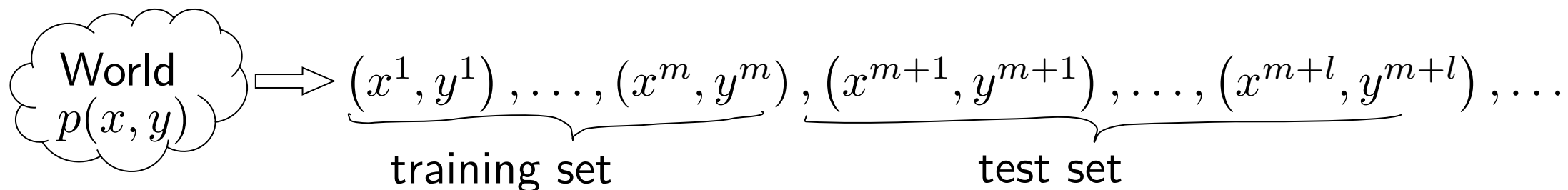
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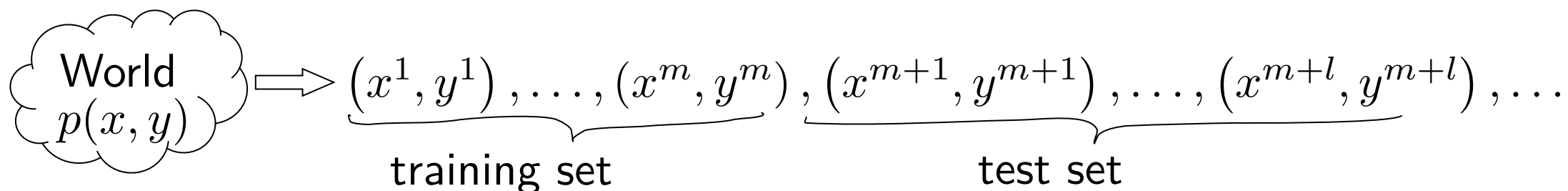


- ◆ **Learning:** find a predictor $h: \mathcal{X} \rightarrow \mathcal{Y}$ with small expected risk $R(h)$ using **training (sequence) set**

$$\mathcal{T}^m = ((x^i, y^i) \in (\mathcal{X} \times \mathcal{Y}) \mid i = 1, \dots, m) \quad \text{drawn i.i.d. from } p(x, y)$$

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- ◆ **Evaluation:** estimate the expected risk $R(h)$ of a given predictor $h: \mathcal{X} \rightarrow \mathcal{Y}$ using **test (sequence) set**

$$\mathcal{S}^l = ((x^i, y^i) \in (\mathcal{X} \times \mathcal{Y}) \mid i = 1, \dots, l) \quad \text{drawn i.i.d. from } p(x, y)$$

Evaluation

- ◆ **Goal:** Given a predictor $h: \mathcal{X} \rightarrow \mathcal{Y}$, estimate the expected risk $R(h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$
- ◆ **Approach:**
 - Collect a test set $\mathcal{S}^l = ((x^i, y^i) \in \mathcal{X} \times \mathcal{Y} \mid i = 1, \dots, l)$ drawn i.i.d. from $p(x, y)$.
 - Estimate the expected risk $R(h)$ by computing the empirical risk (test error)

$$R_{\mathcal{S}^l}(h) = \frac{1}{l}(\ell(y^1, h(x^1)) + \dots + \ell(y^l, h(x^l))) = \frac{1}{l} \sum_{i=1}^l \ell(y^i, h(x^i))$$

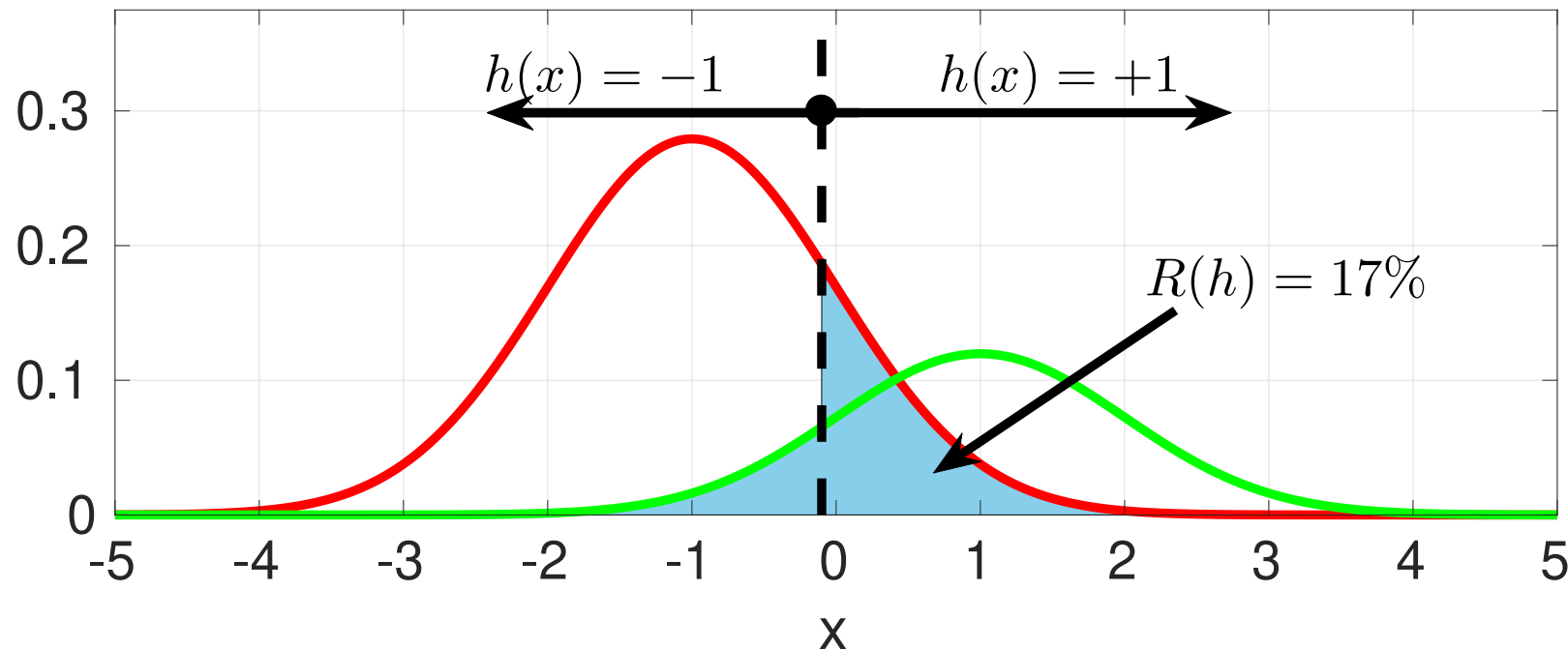
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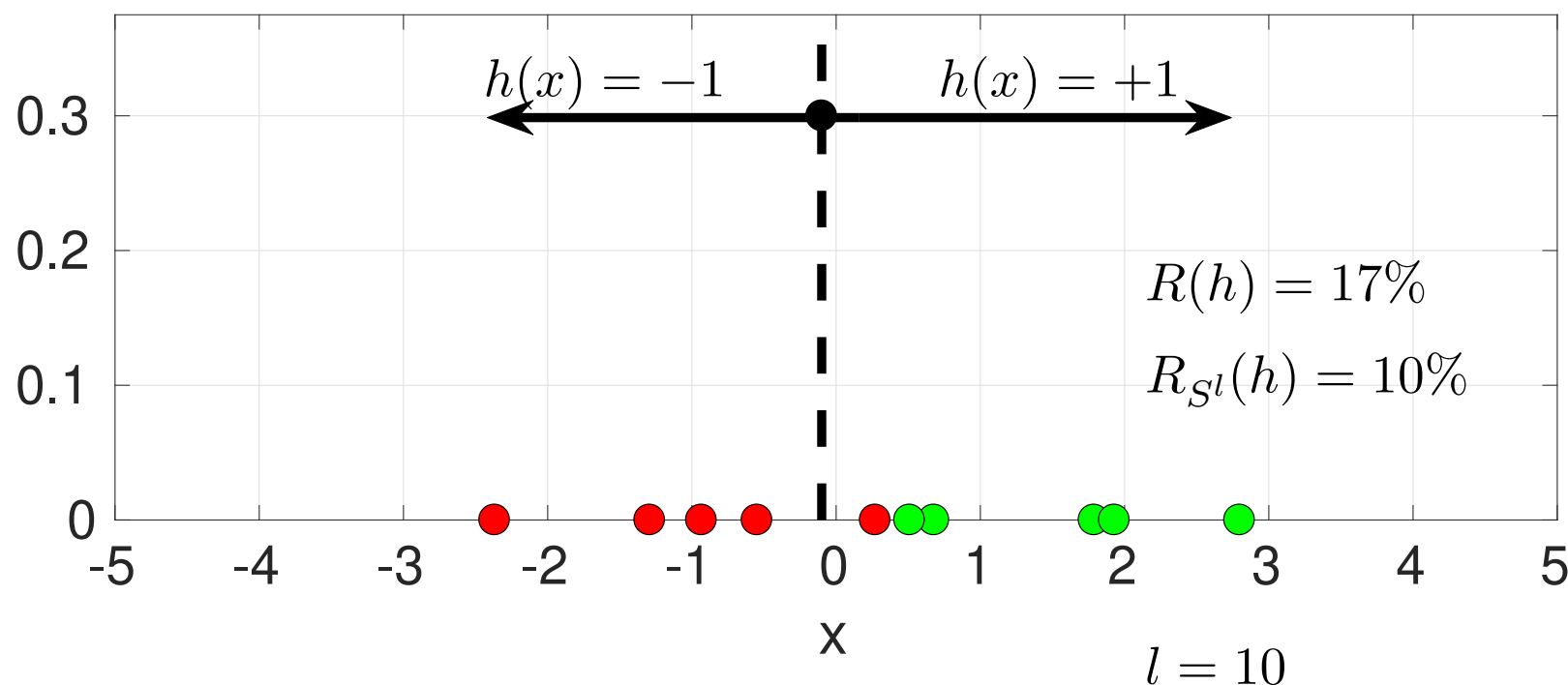
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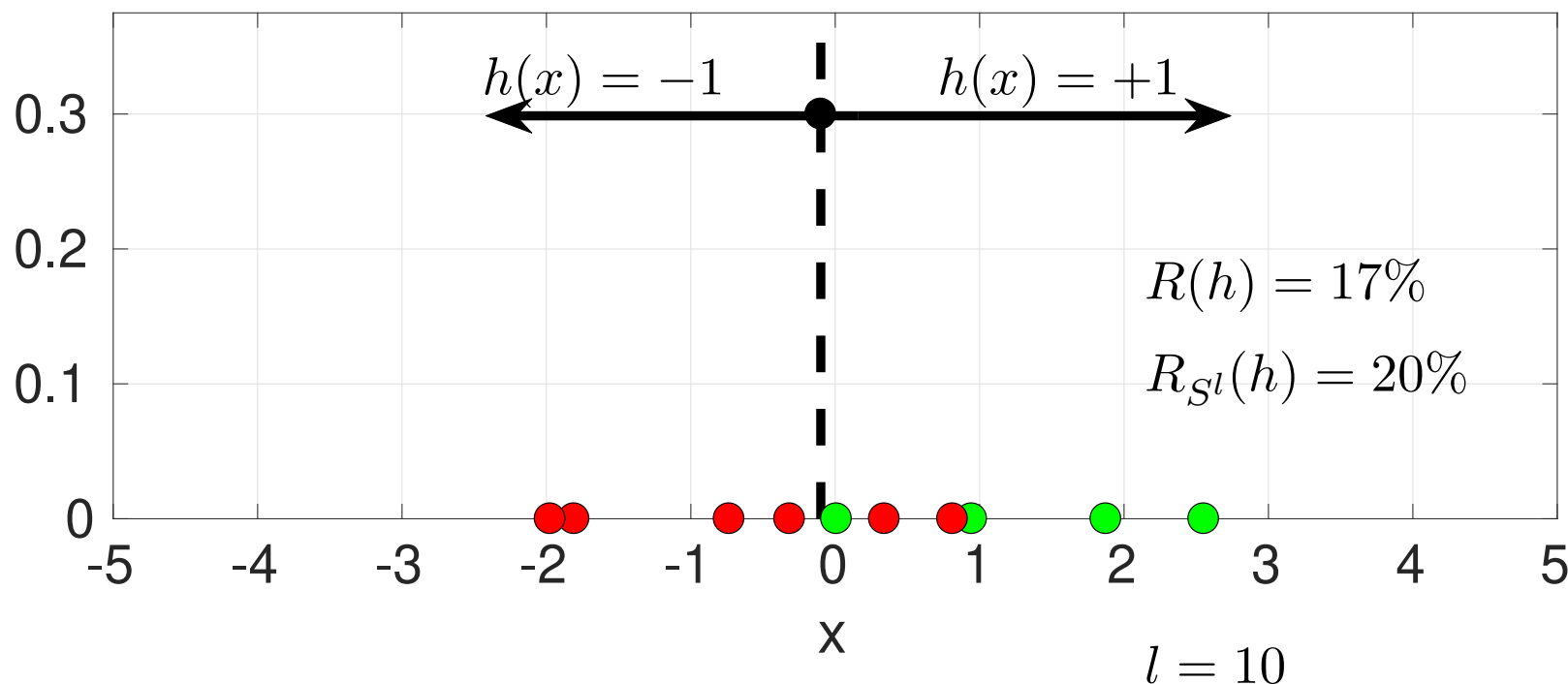
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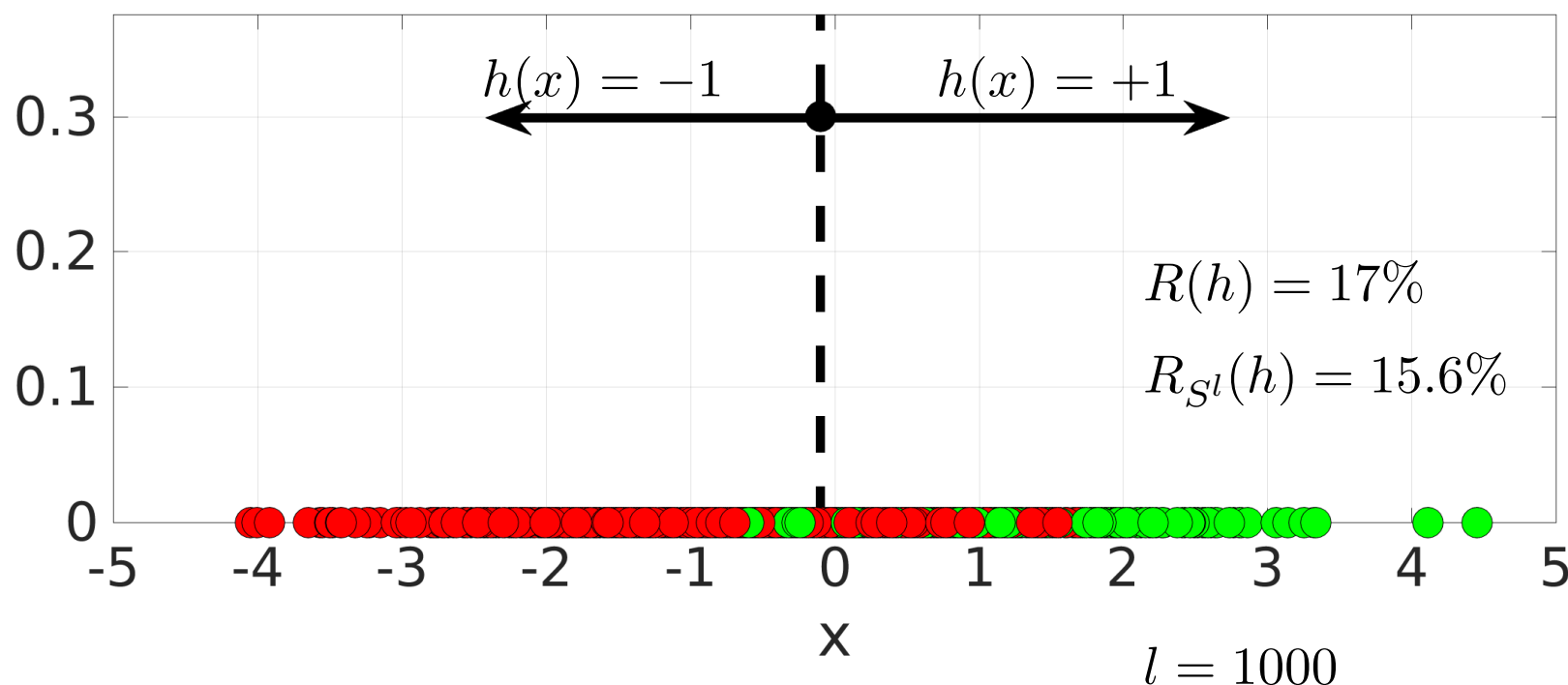
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- ◆ **Properties of $R_{\mathcal{S}^l}(h)$ as an estimator of $R(h)$:**

- $R_{\mathcal{S}^l}(h)$ is a random number with an unknown distribution.
- $R_{\mathcal{S}^l}(h)$ is unbiased estimator of the expected risk $R(h)$, i.e.

$$\mathbb{E}_{\mathcal{S}^l \sim p^l}[R_{\mathcal{S}^l}(h)] = R(h)$$

- We will quantify the deviation between $R(h)$ and $R_{\mathcal{S}^l}(h)$ via a bound:

$$\mathbb{P}_{\mathcal{S}^l \sim p^l}(|R_{\mathcal{S}^l}(h) - R(h)| \geq \varepsilon) \leq \text{Function of } l \text{ and } \ell$$

Law of large numbers

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$$z^2 = 1$$



$$z^3 = 5$$

...



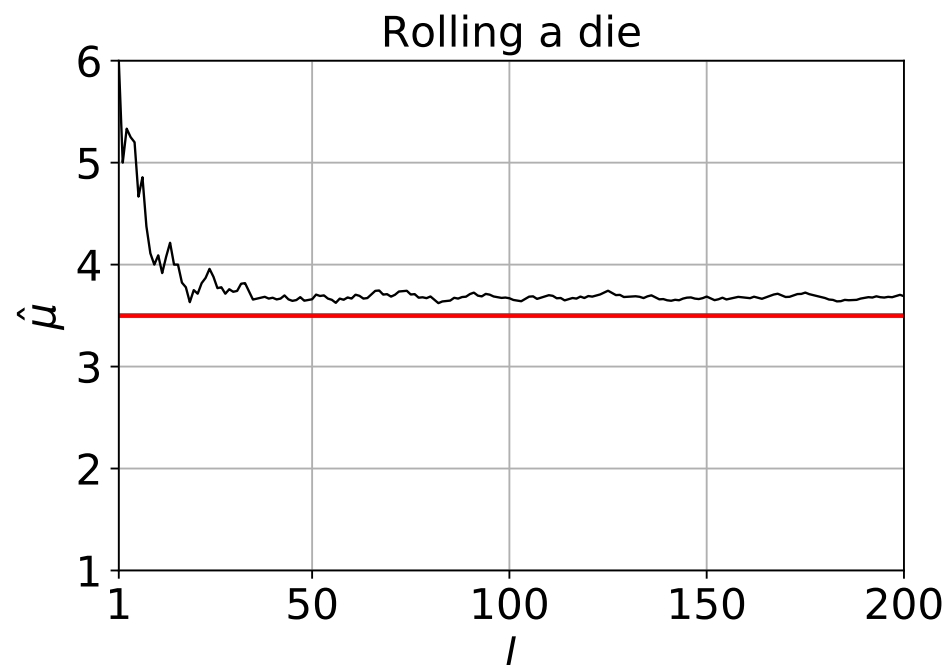
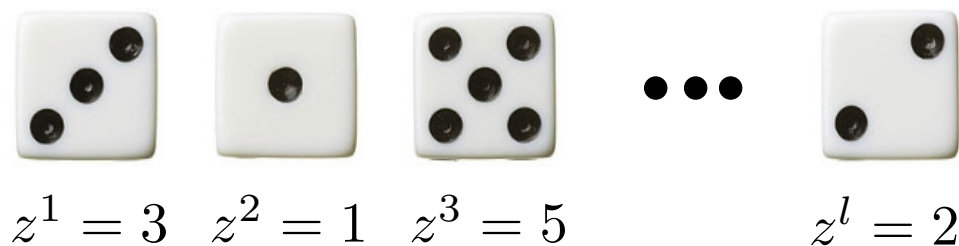
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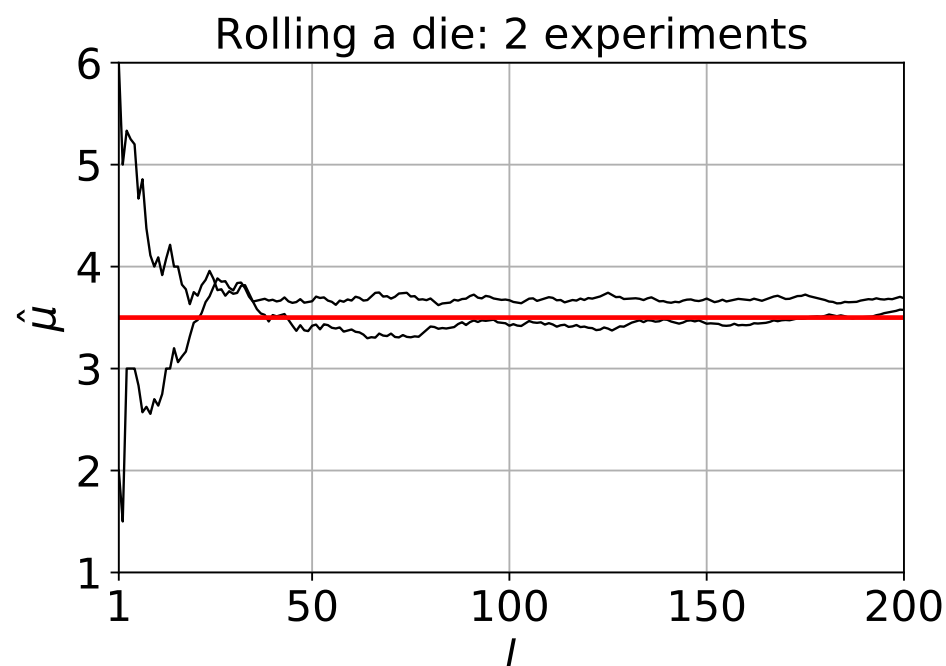


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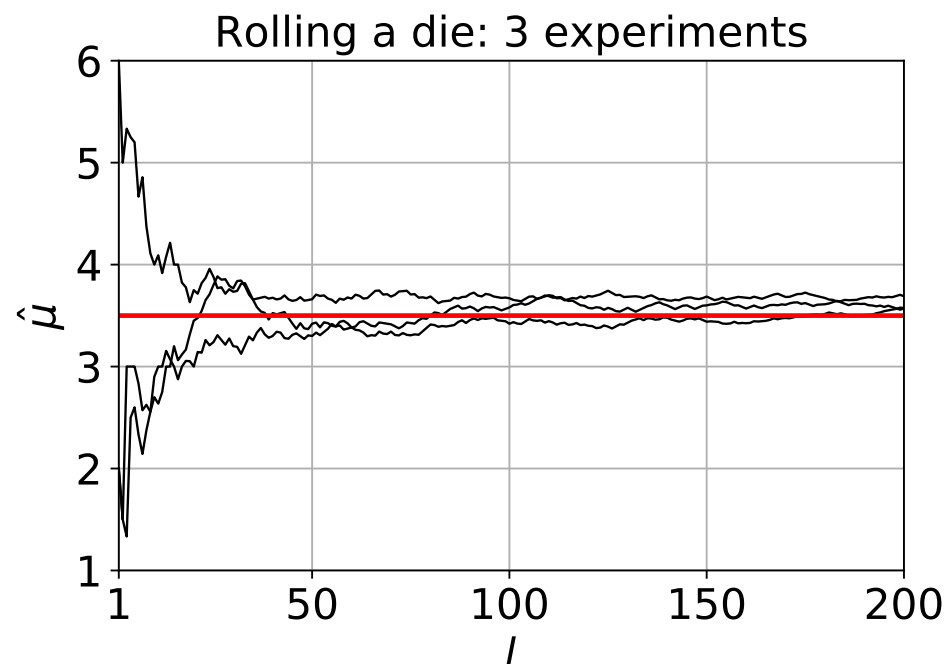
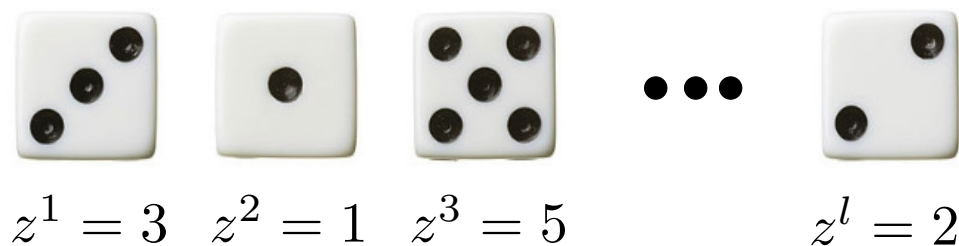


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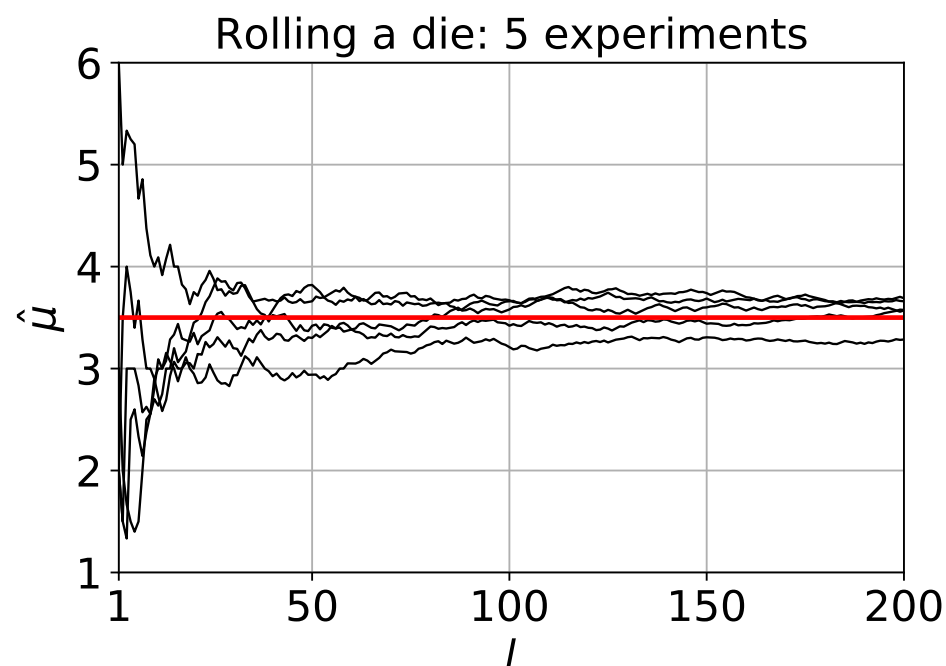
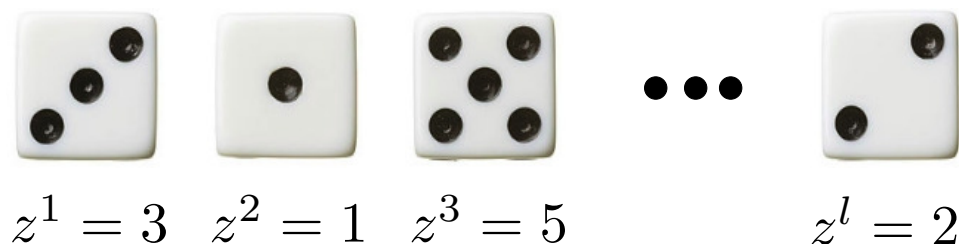


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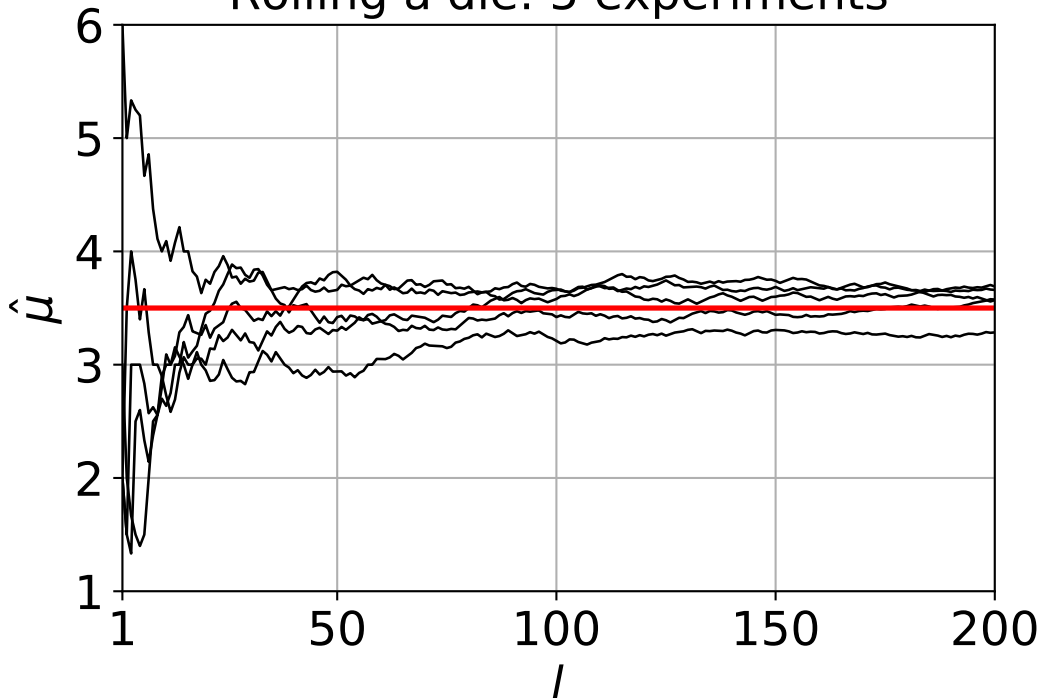
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Given a finite sample size l , how effectively does the sample mean $\hat{\mu}_l$ estimate the expected value μ ?

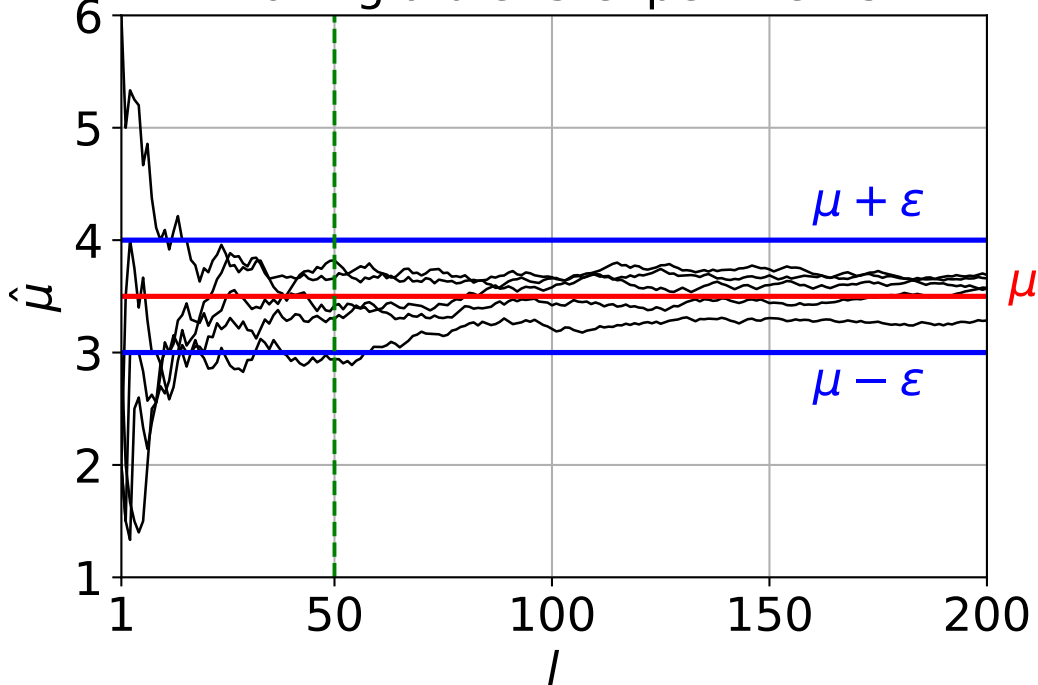
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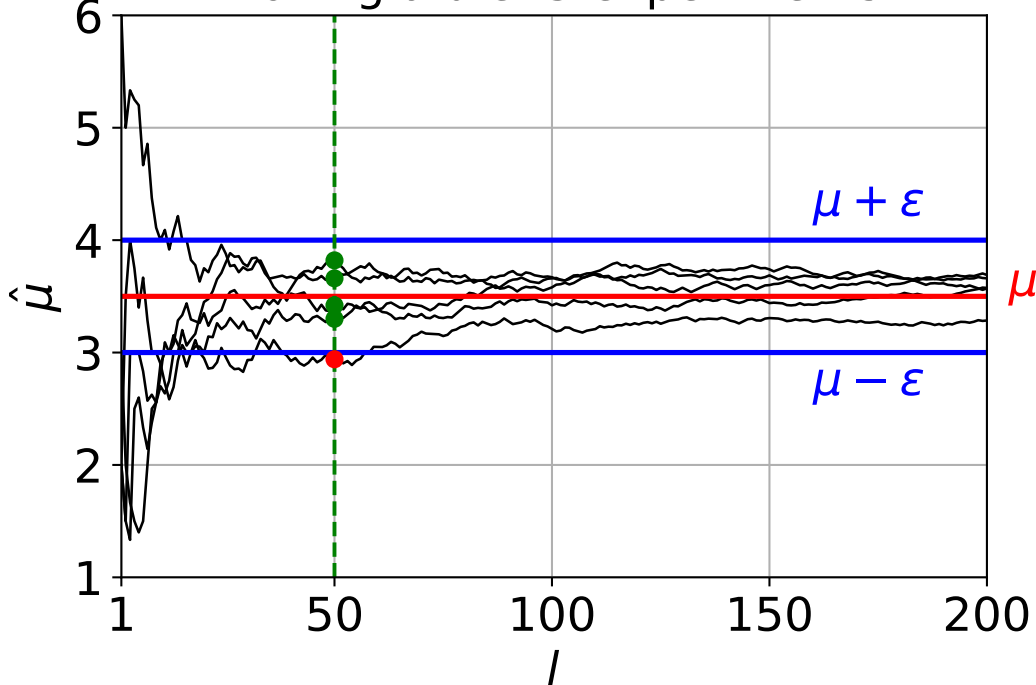
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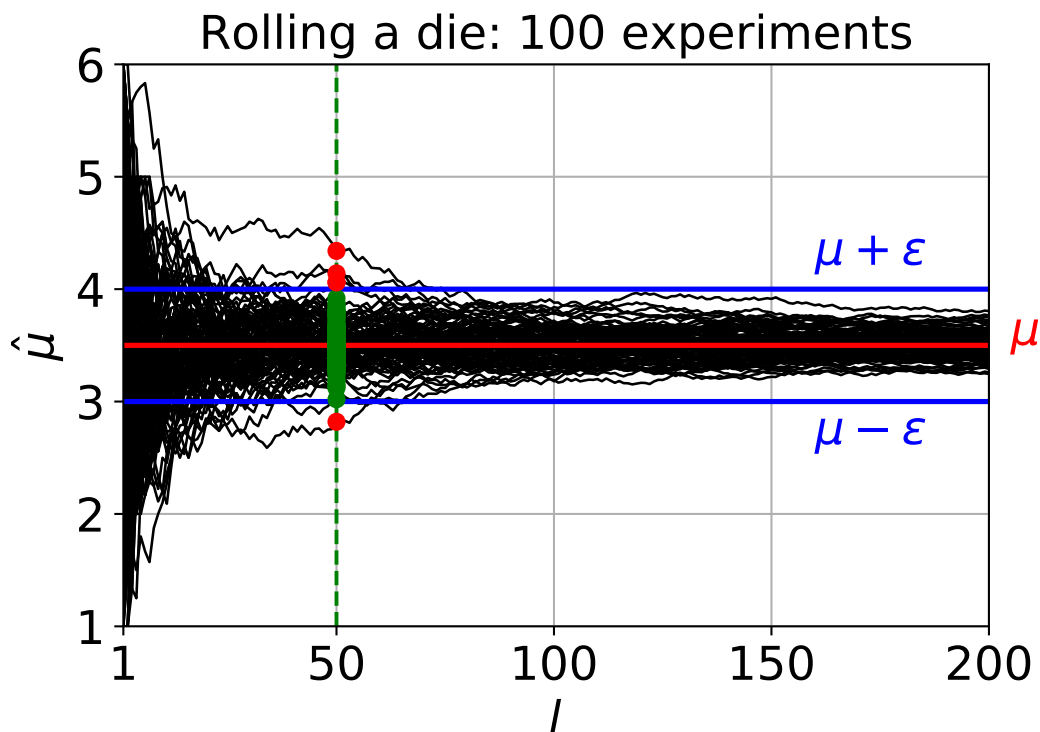
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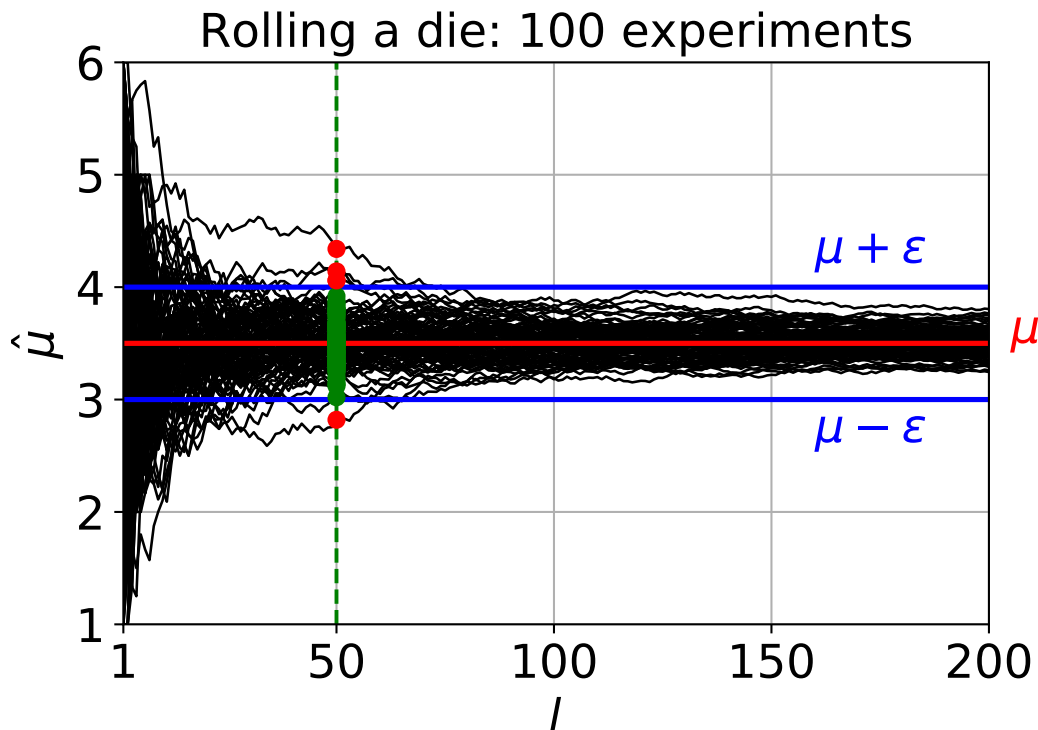
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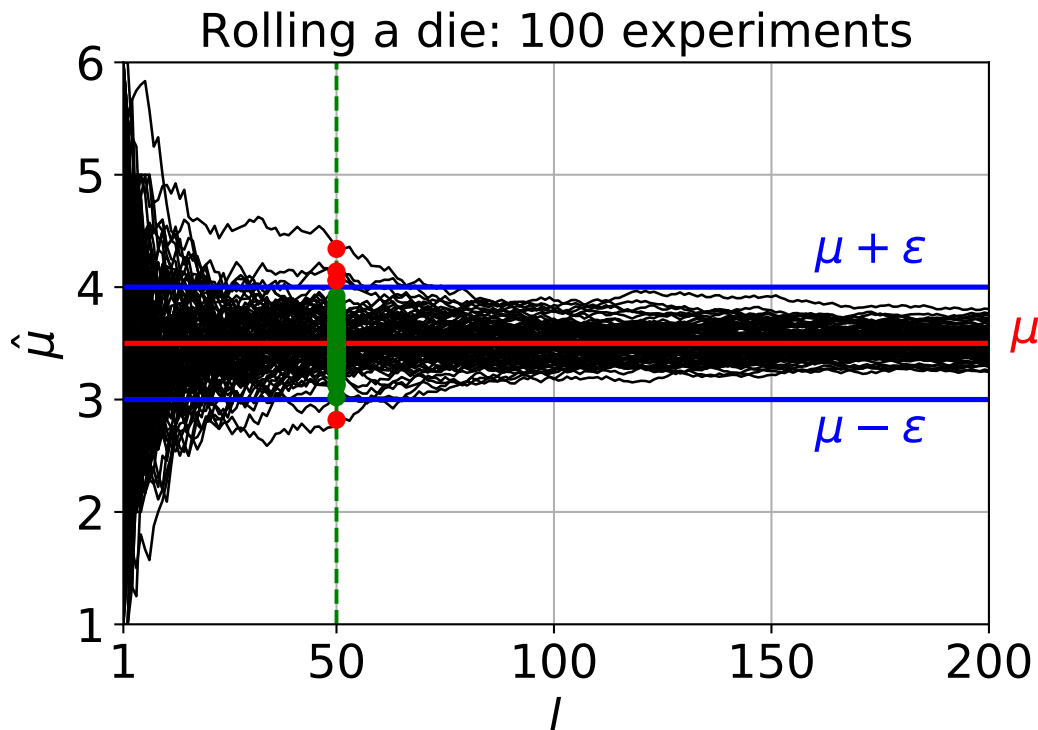
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$$\frac{\#(|\hat{\mu}_l - \mu| \geq \varepsilon)}{\#\text{experiments}} = \frac{5}{100} = 0.05 \quad \rightarrow \quad \mathbb{P}_{\mathcal{Z}^l \sim q^l}(|\hat{\mu}_l - \mu| \geq \varepsilon)$$

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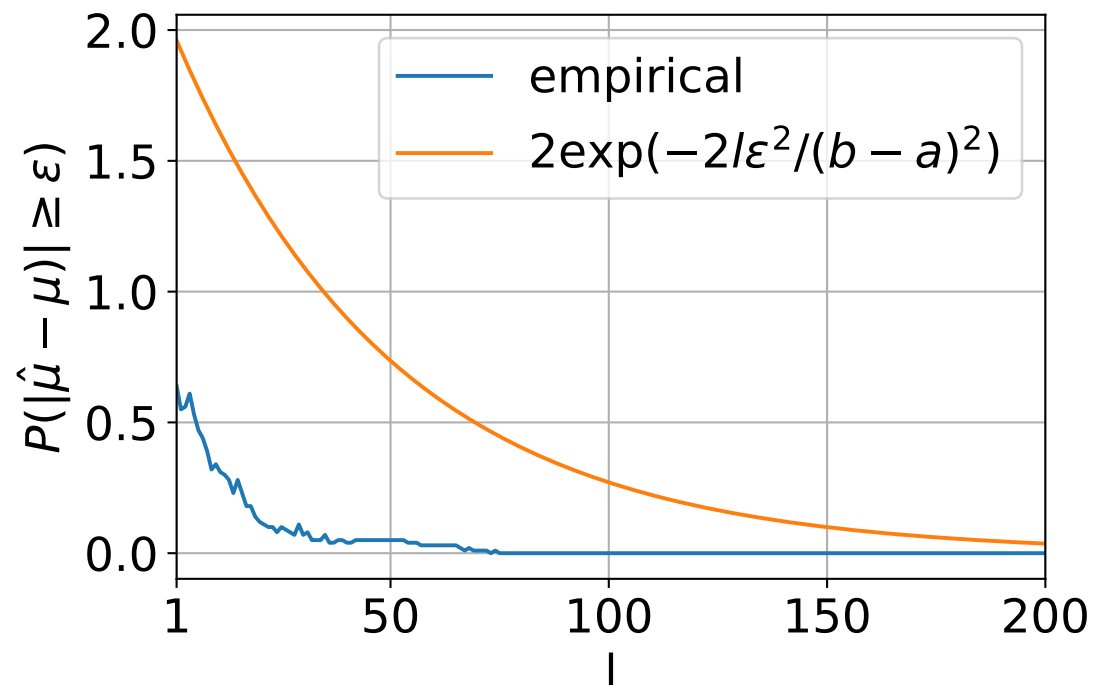
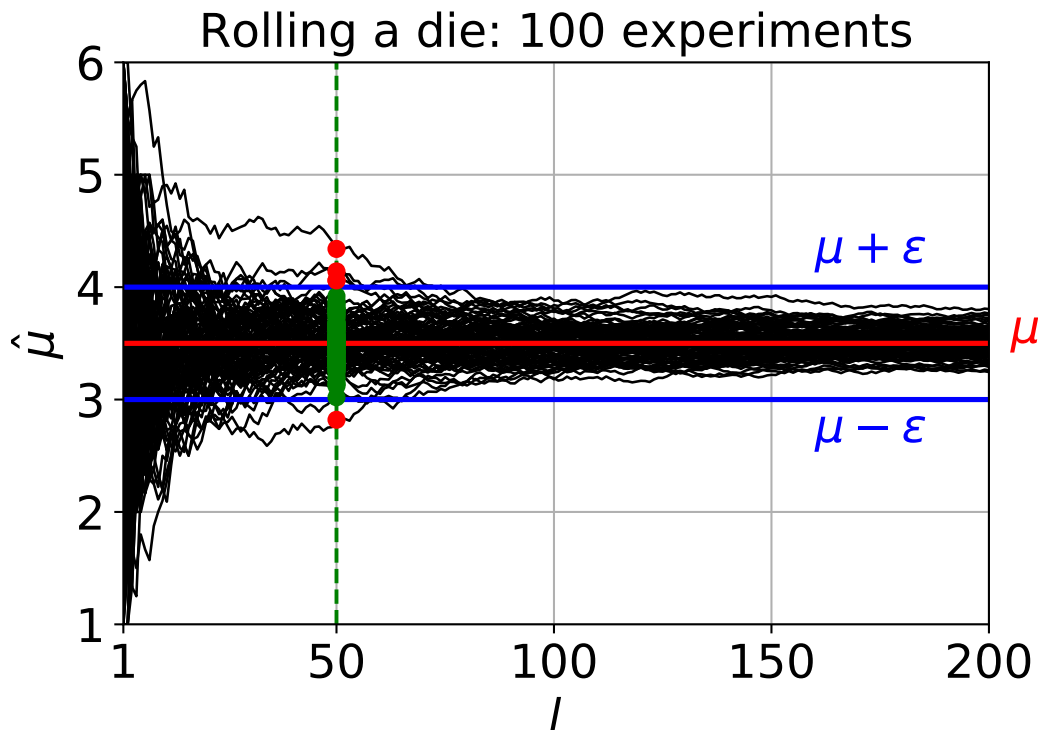
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Hoeffding inequality

$$\frac{\#(|\hat{\mu}_l - \mu| \geq \varepsilon)}{\#\text{experiments}} = \frac{5}{100} = 0.05 \quad \rightarrow \quad \mathbb{P}_{\mathcal{Z}^l \sim q^l}(|\hat{\mu}_l - \mu| \geq \varepsilon) \leq 2e^{-\frac{2l\varepsilon^2}{(b-a)^2}}$$

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- ◆ **Theorem:** Let $\mathcal{Z}^l = (z^1, \dots, z^l)$ be i.i.d. sample generated from r.v. with distribution $q(z)$. Let the random variables attain values from an interval $[a, b]$. Let the expected value of the r.v. be $\mu = \mathbb{E}_{z \sim q}(z)$. Let $\hat{\mu}_l = \frac{1}{l} \sum_{i=1}^l z^i$. Then for any $\varepsilon > 0$ it holds that

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◆ **Properties of the Hoeffding inequality:**

- (-) Conservative: the bound may not be tight.
- (+) General: the bound holds for any distribution over bounded r.v.
- (+) Cheap: The bound is simple and easy to compute.

Applying Hoeffding to the expected risk estimation

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- ◆ **Our approach:** Given $h: \mathcal{X} \rightarrow \mathcal{Y}$ and test set $\mathcal{S}^l \sim p^l$, we estimate the expected risk $R(h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$ by the empirical risk $R_{\mathcal{S}^l}(h) = \frac{1}{l} \sum_{i=1}^l \ell(y^i, h(x^i))$.

- ◆ **Translation:**

$$z^i = \ell(y^i, h(x^i)) \quad \hat{\mu}_l = R_{\mathcal{S}^l}(h) \quad \mu = R(h)$$

The Hoeffding inequality applied to the expected risk estimation:

$$\mathbb{P}_{\mathcal{S}^l \sim p^l} \left(|R_{\mathcal{S}^l}(h) - R(h)| \geq \varepsilon \right) \leq 2e^{-\frac{2l\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}} \quad \forall \varepsilon > 0$$

Confidence interval for the expected risk

- ◆ **Goal:** Characterize the deviation between the expected risk $R(h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$ and the empirical risk $R_{\mathcal{S}^l}(h) = \frac{1}{l} \sum_{i=1}^l \ell(y^i, h(x^i))$ computed on the test set $\mathcal{S}^l \sim p^l$.

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- ◆ **Hoeffding inequality:** provides a probabilistic bound on the deviation between the expected risk $R(h)$ and the empirical risk $R_{\mathcal{S}^l}(h)$:

$$\mathbb{P}_{\mathcal{S}^l \sim p^l} \left(|R(h) - R_{\mathcal{S}^l}(h)| \geq \varepsilon \right) \leq 2e^{-\frac{2l\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}} \quad \forall \varepsilon > 0$$

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- ◆ **Remark:** The confidence interval is valid for any data distribution.

Learning: formal definition

- ◆ **Goal:** Given a training set $\mathcal{T}^m \sim p^m$, find a strategy $h: \mathcal{X} \rightarrow \mathcal{Y}$ minimizing the expected risk $R(h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$.

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- ◆ **Learning algorithm:** is a function

$$A: \cup_{m=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^m \rightarrow \mathcal{H}$$

returns a strategy $h_m = A(\mathcal{T}^m)$ from $\mathcal{H}$ based on a training set $\mathcal{T}^m$

Empirical Risk Minimization approach to learning

- ◆ The unknown expected risk $R(h)$ is approximated by the empirical risk:

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which can be computed the training set $\mathcal{T}^m = ((x^i, y^i) \in \mathcal{X} \times \mathcal{Y} \mid i = 1, \dots, m) \sim p^m$.

- ◆ The ERM based learning algorithm returns h_m such that

$$h_m \in \underset{h \in \mathcal{H}}{\text{Argmin}} R_{\mathcal{T}^m}(h) \quad (1)$$

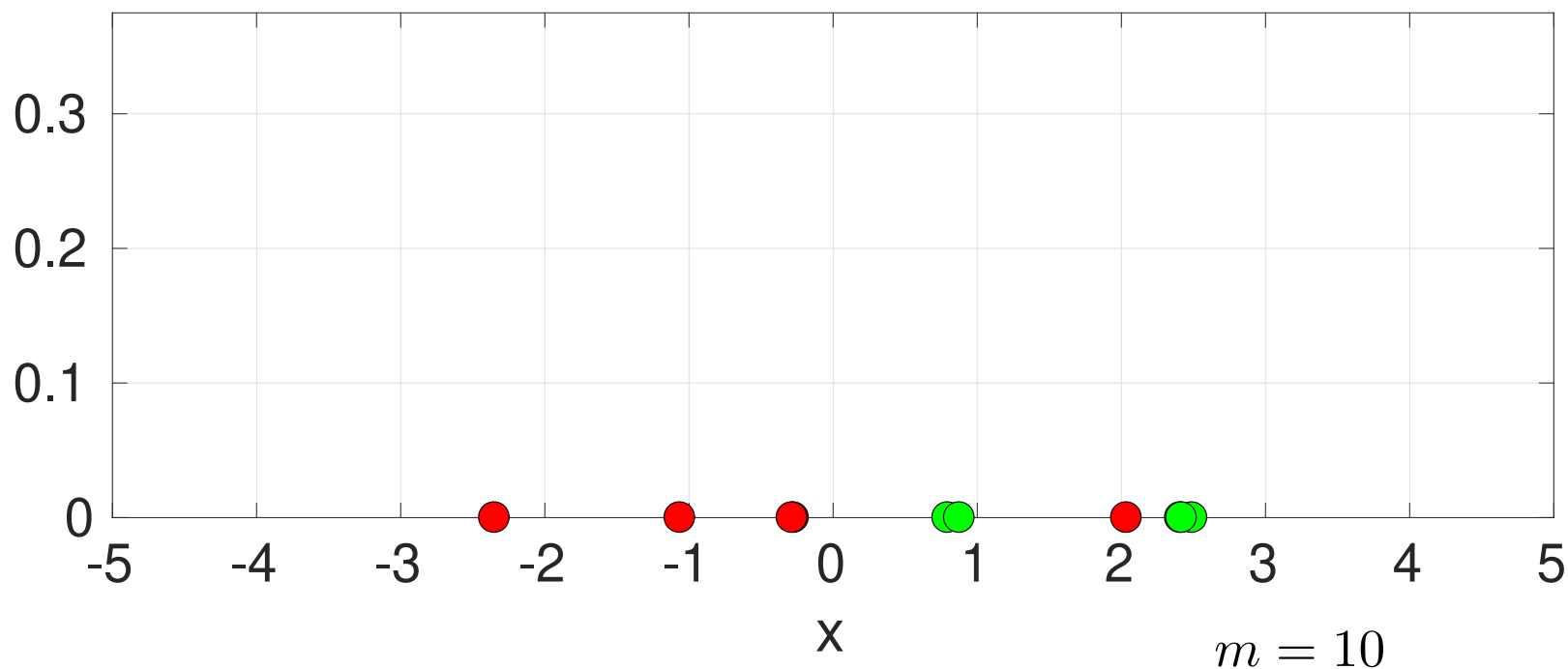
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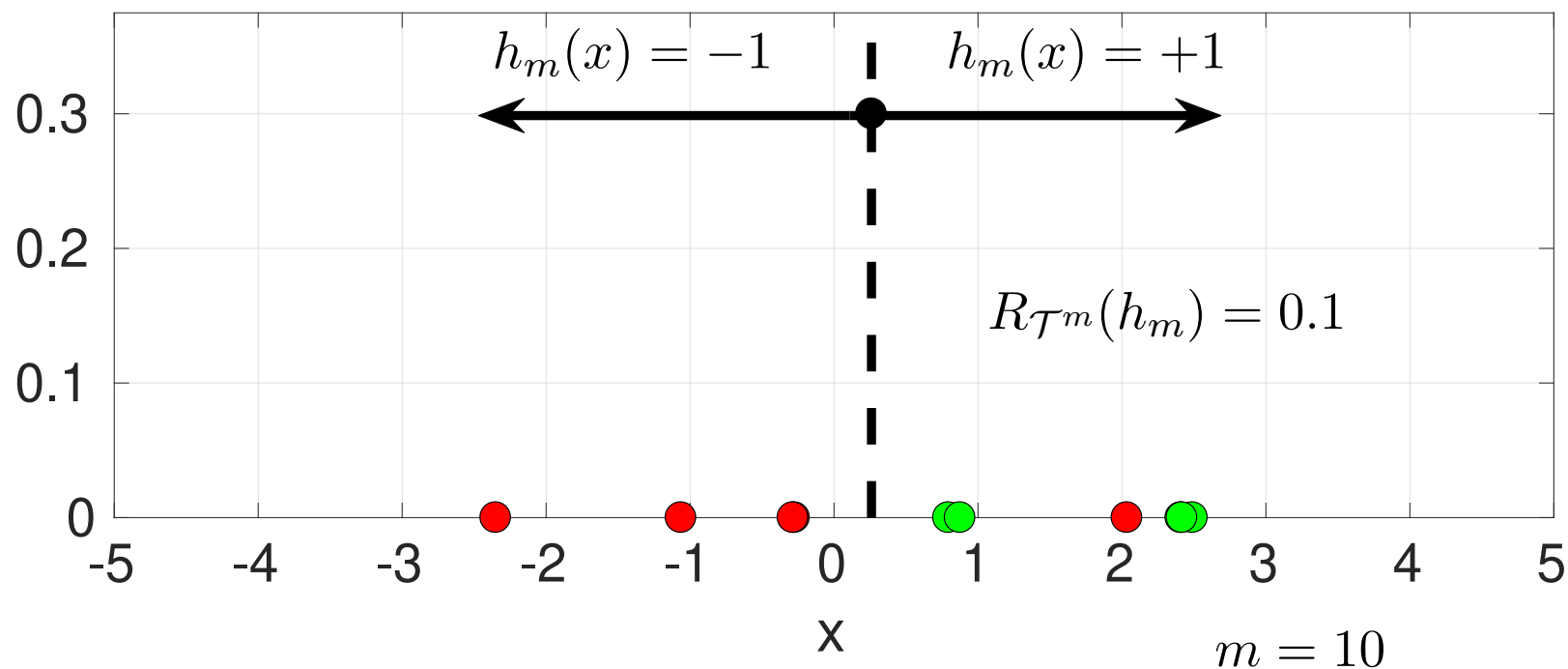
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- ◆ Depending on the choince of $\mathcal{H}$ and ℓ and algorithm solving (1) we get individual instances e.g. Support Vector Machines, Linear Regression, Logistic Regression, Neural Networks learned by back-propagation, AdaBoost, Gradient Boosted Trees, ...

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- ◆ Let $\mathcal{X} = [a, b] \subset \mathbb{R}$, $\mathcal{Y} = \{+1, -1\}$, $\ell(y, y') = \mathbb{I}[y \neq y']$, $p(x \mid y = +1)$ and $p(x \mid y = -1)$ be uniform distributions on $\mathcal{X}$ and $p(y = +1) = 0.8$.

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- ◆ **Next lecture:** we will show that overfitting can be eliminated by constraining the hypothesis space $\mathcal{H}$, i.e. by using a prior knowledge.

Why the law of large numbers does not apply for learning?

◆ Hoeffding inequality: (as an instance of LLN)

- $\mathbb{P}(|\hat{\mu} - \mu| \geq \varepsilon) \leq 2e^{-\frac{2m\varepsilon^2}{(b-a)^2}}$, where $\hat{\mu} = \frac{1}{m} \sum_{i=1}^m z^i$ assumes that $(z^1, \dots, z^m)$ is i.i.d. sample generated from rand. vars. with the expected value μ .

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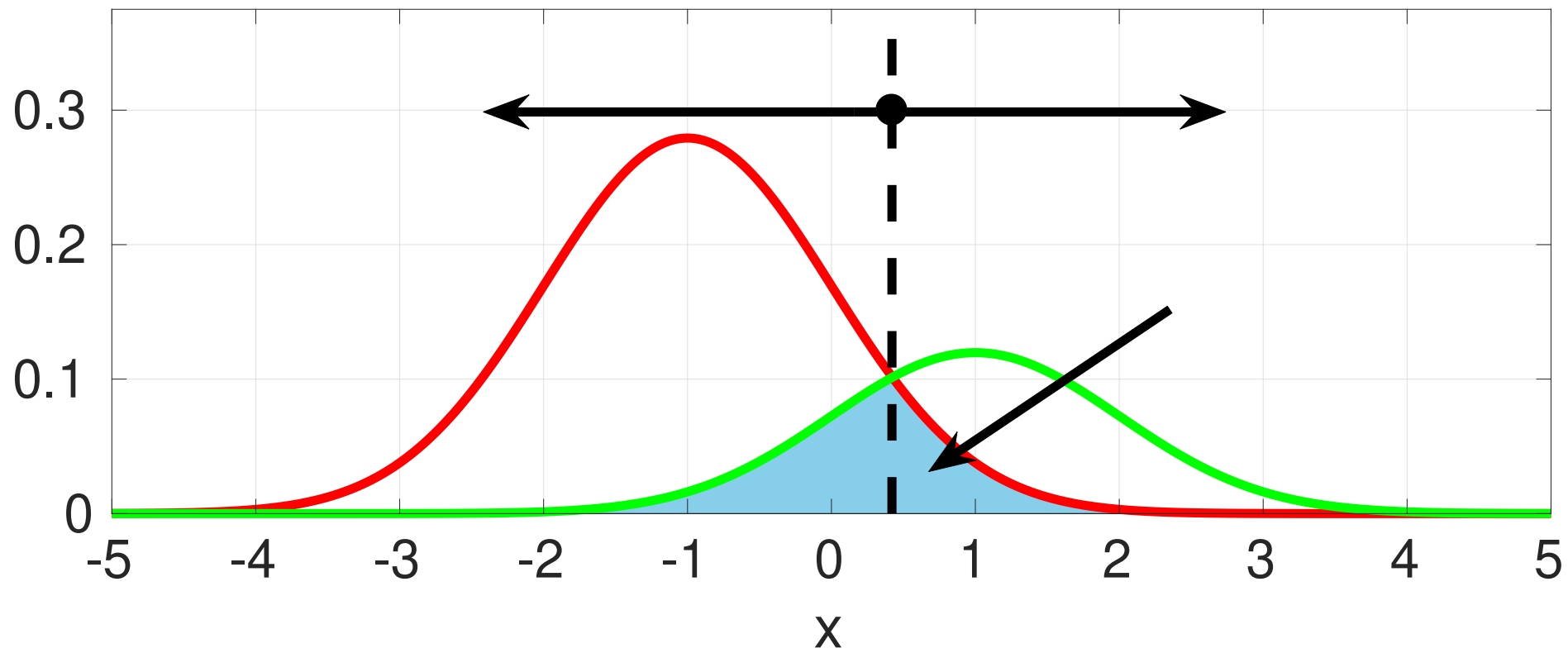
◆ Learning:

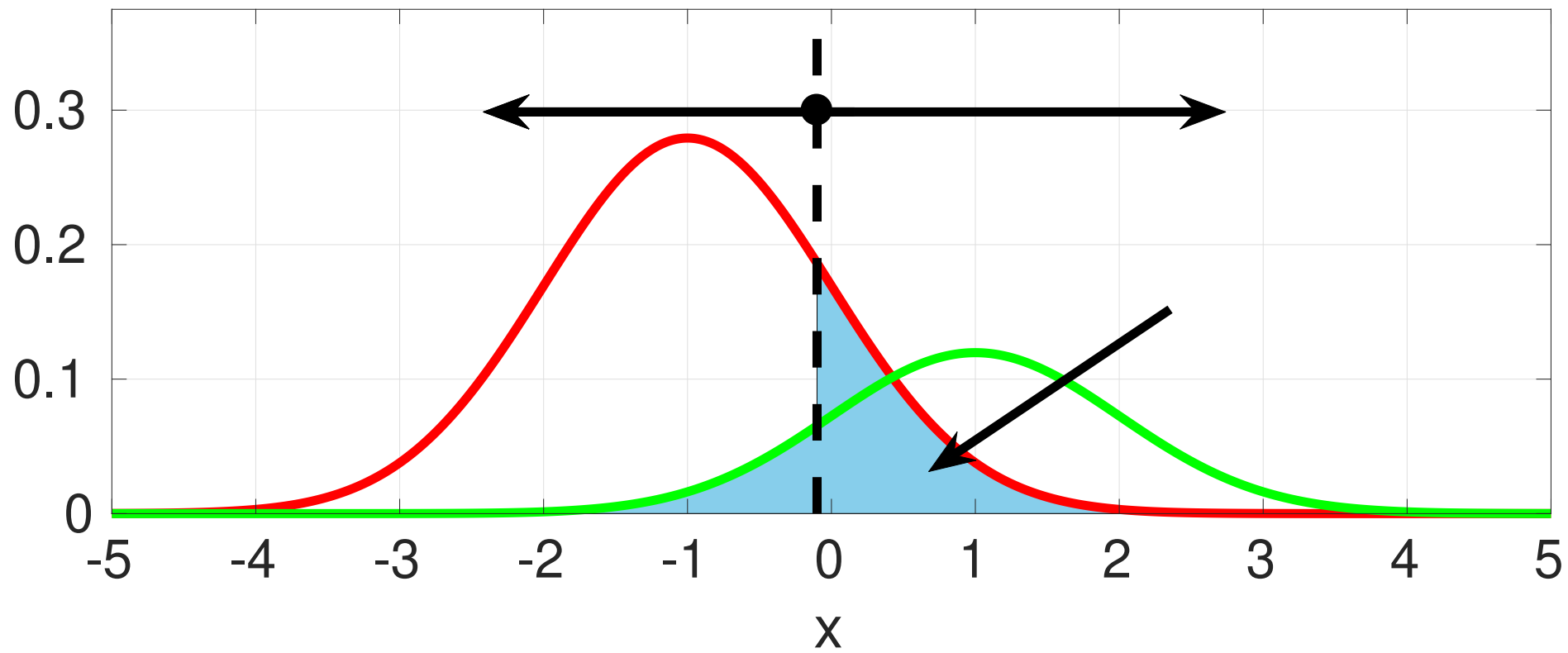
- $h_m = A(\mathcal{T}^m)$, $z^i = \ell(y^i, h_m(x^i))$, i.e. every z^i is a function of $\mathcal{T}^m$, and thus $(z^1, \dots, z^m)$ is not i.i.d. sample
- We cannot apply Hoeffding to bound $\mathbb{P}(|R_{\mathcal{T}^m}(h_m) - R(h_m)| \geq \varepsilon)$

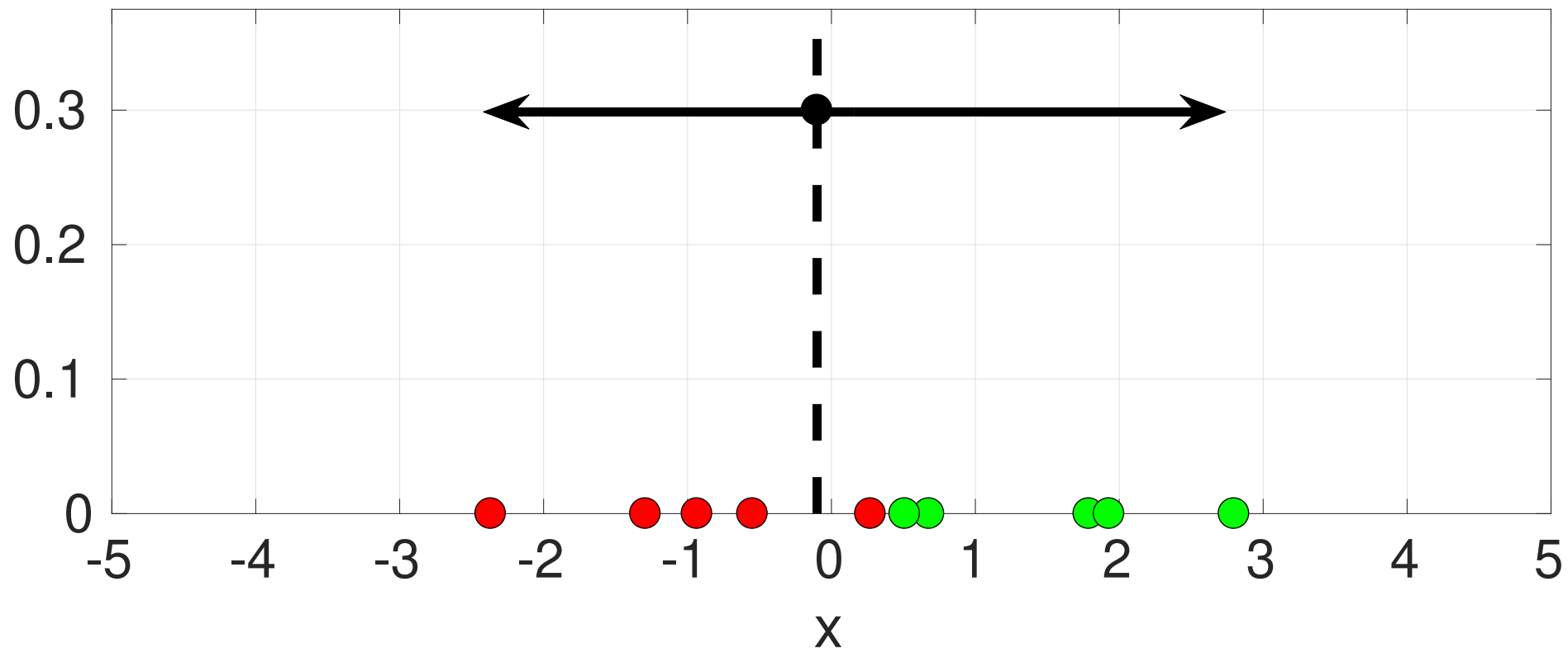
◆ Next lecture: We show how to fix the problem via using the Uniform Law of Large Numbers.

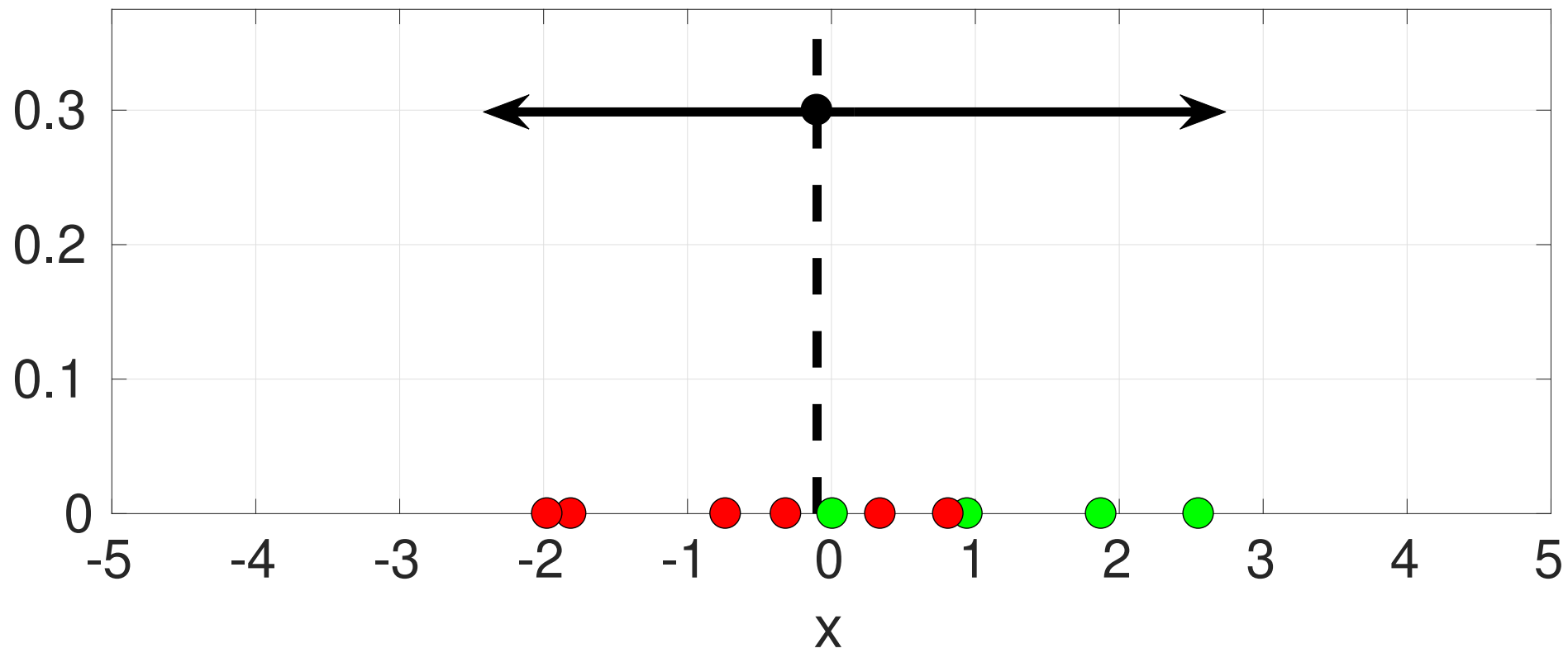
Summary

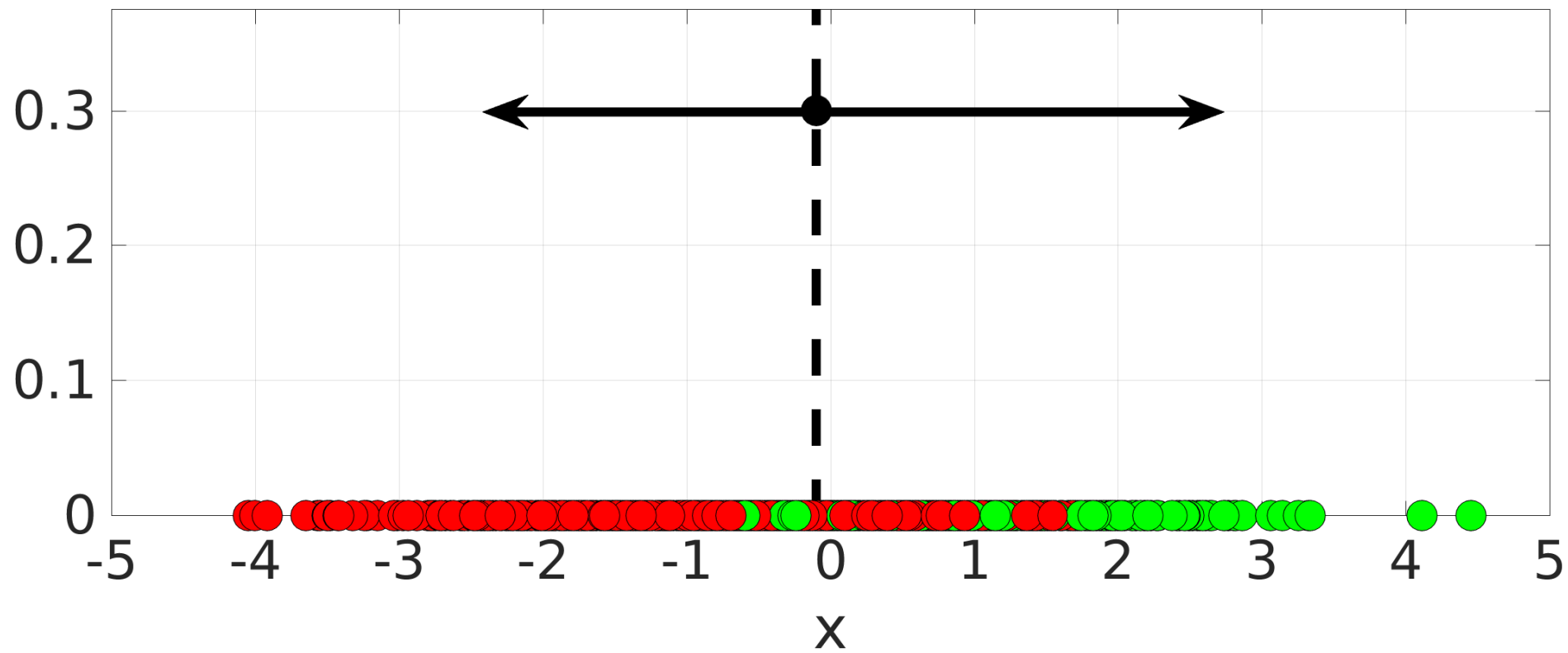
- ◆ Formulation of the prediction problem.
- ◆ Estimation of the expected risk by the empirical risk.
- ◆ Law of Large numbers.
- ◆ Hoeffding inequality.
- ◆ Confidence intervals.
- ◆ Formulation of the learning algorithm.
- ◆ Empirical Risk Minimization.
- ◆ Example demonstrating that ERM can fail due to overfitting.













...



$$z^1 = 3 \quad z^2 = 1 \quad z^3 = 5$$

$$z^l = 2$$



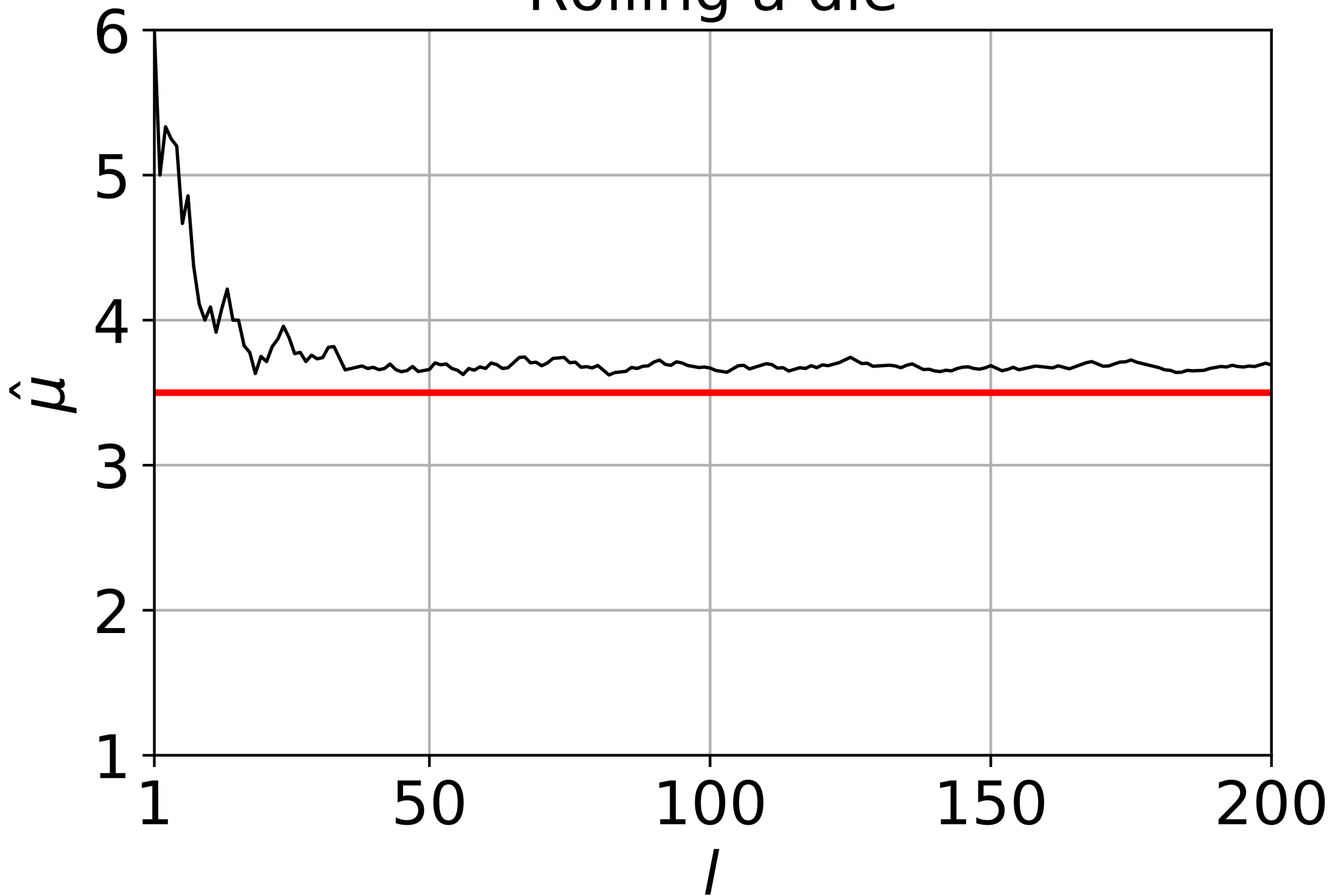
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Rolling a die





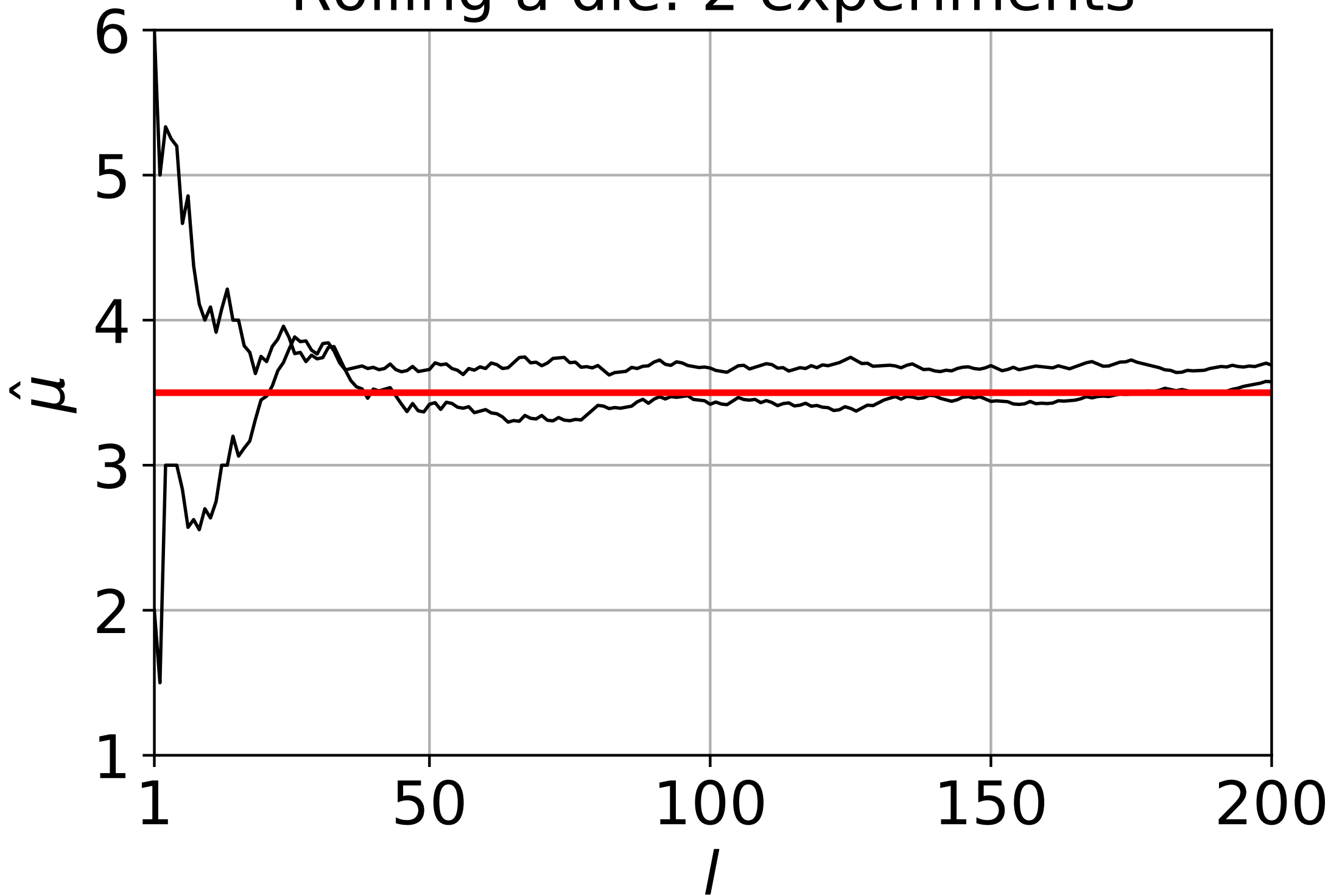
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Rolling a die: 2 experiments





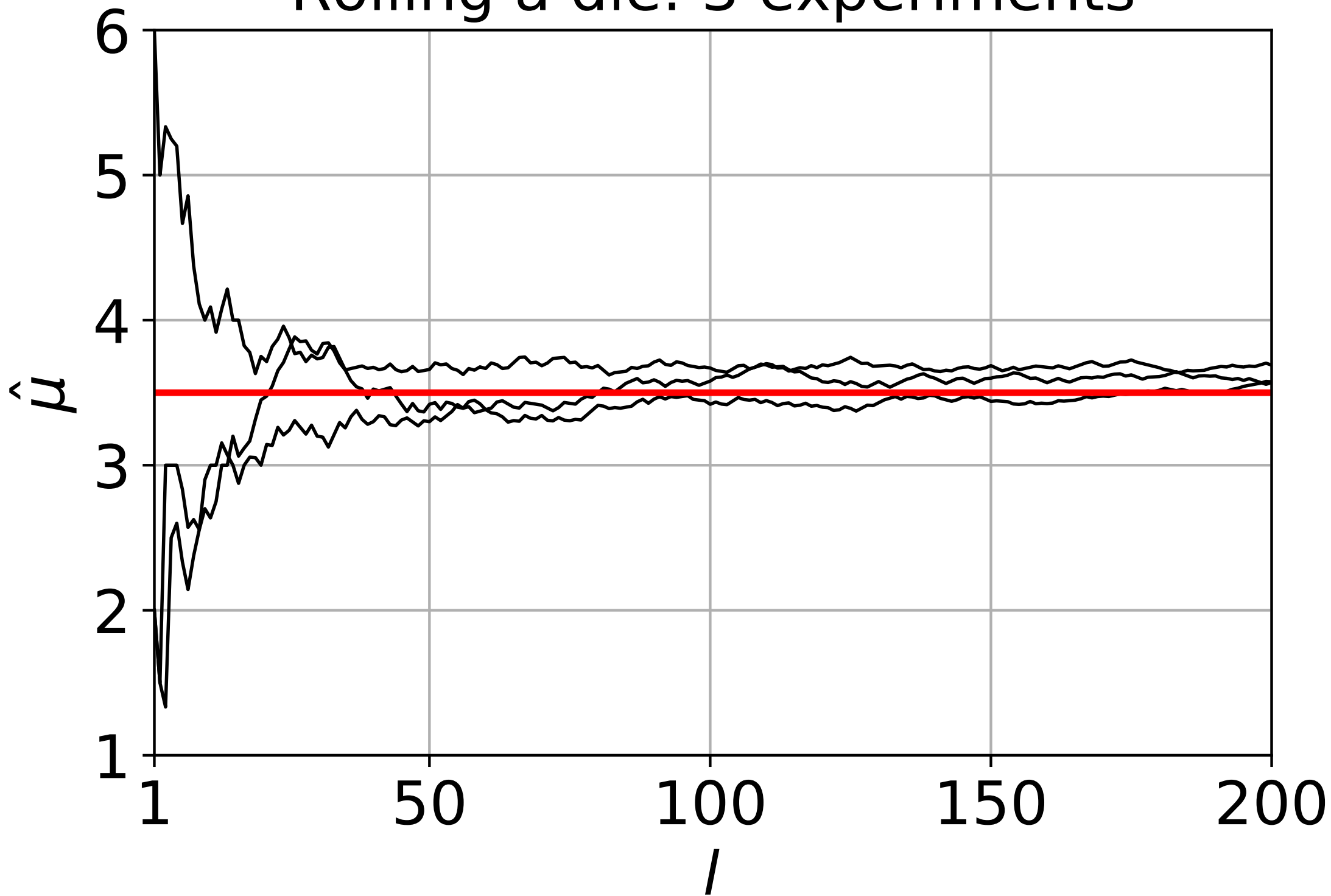
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Rolling a die: 3 experiments





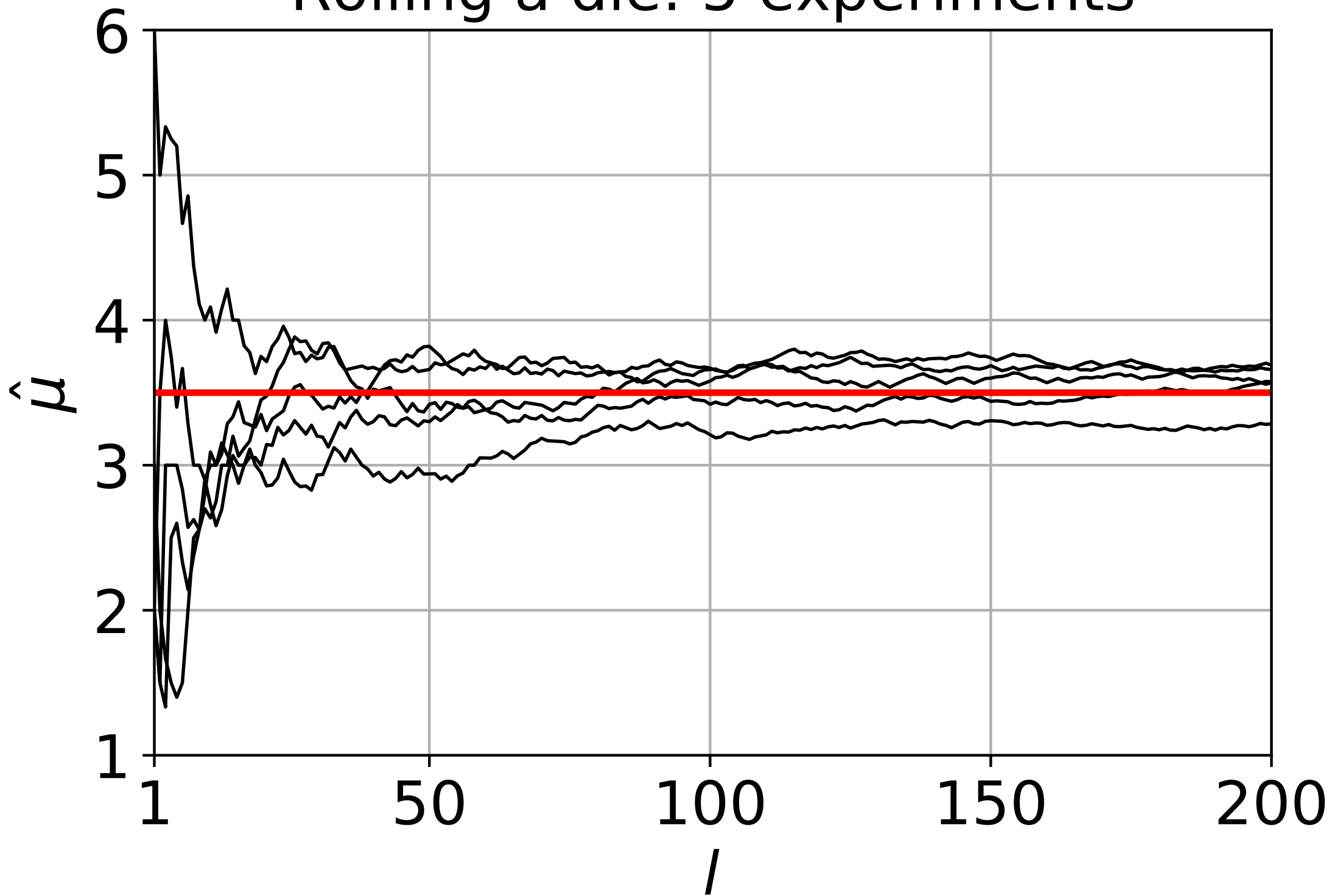
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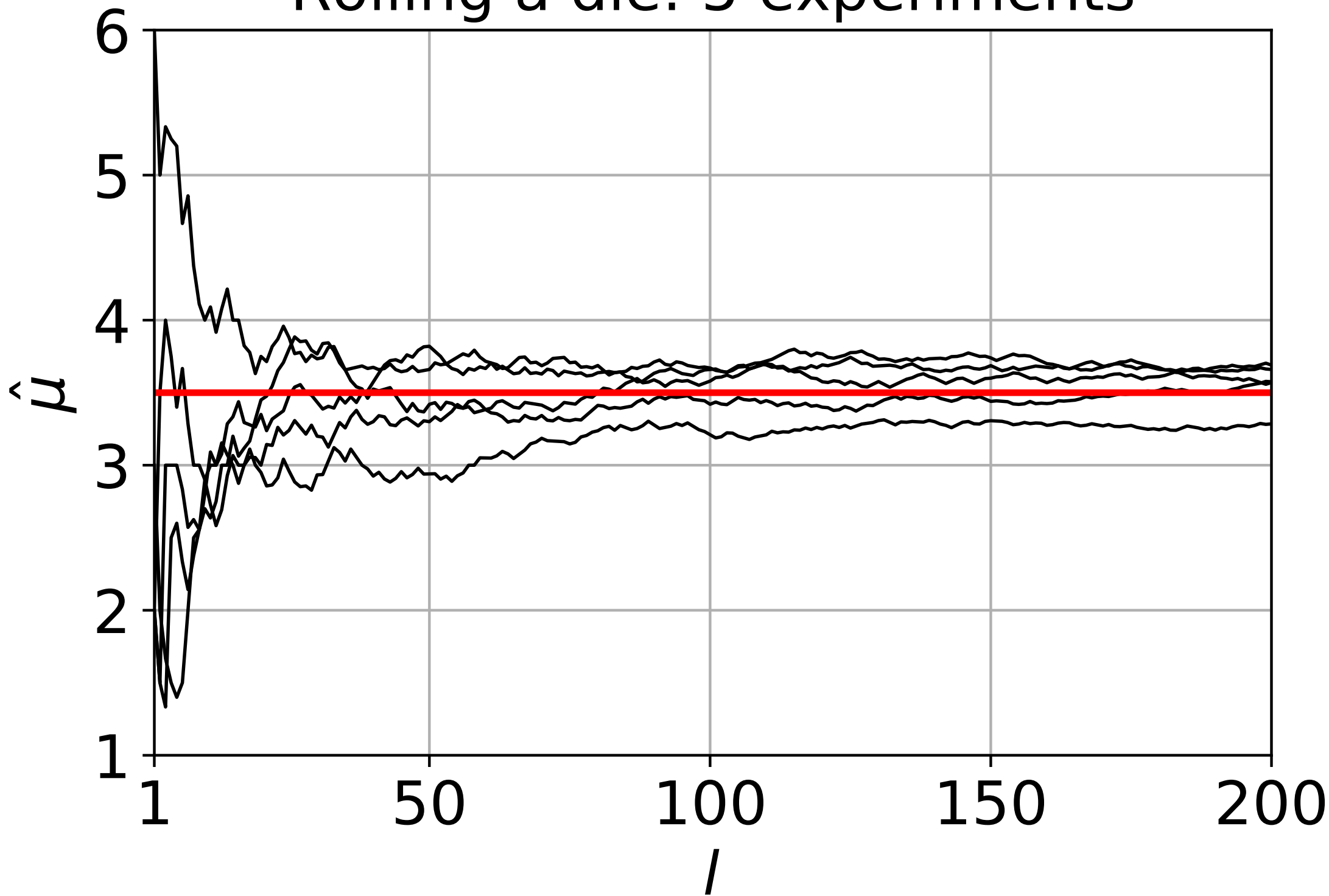
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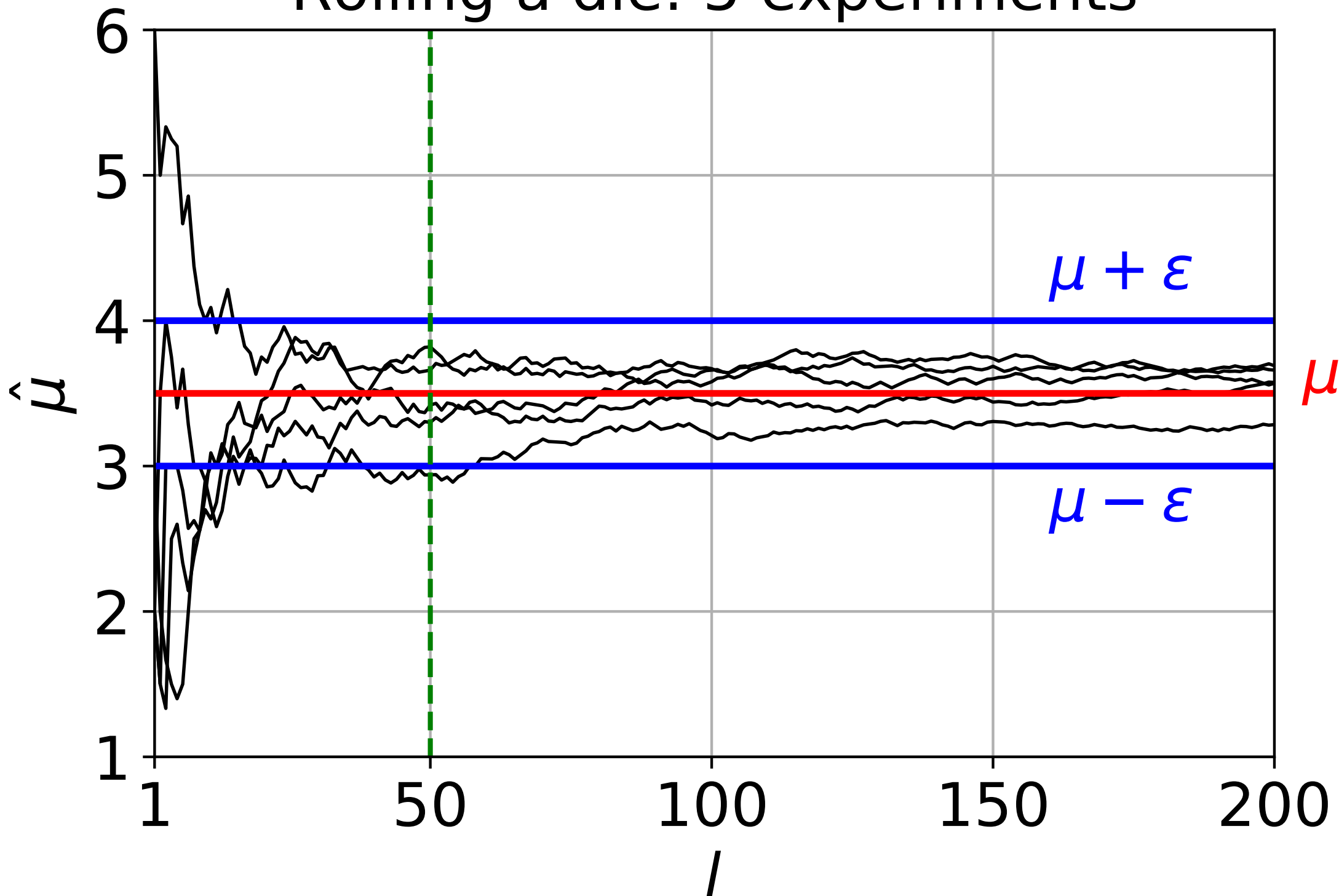
Rolling a die: 5 experiments



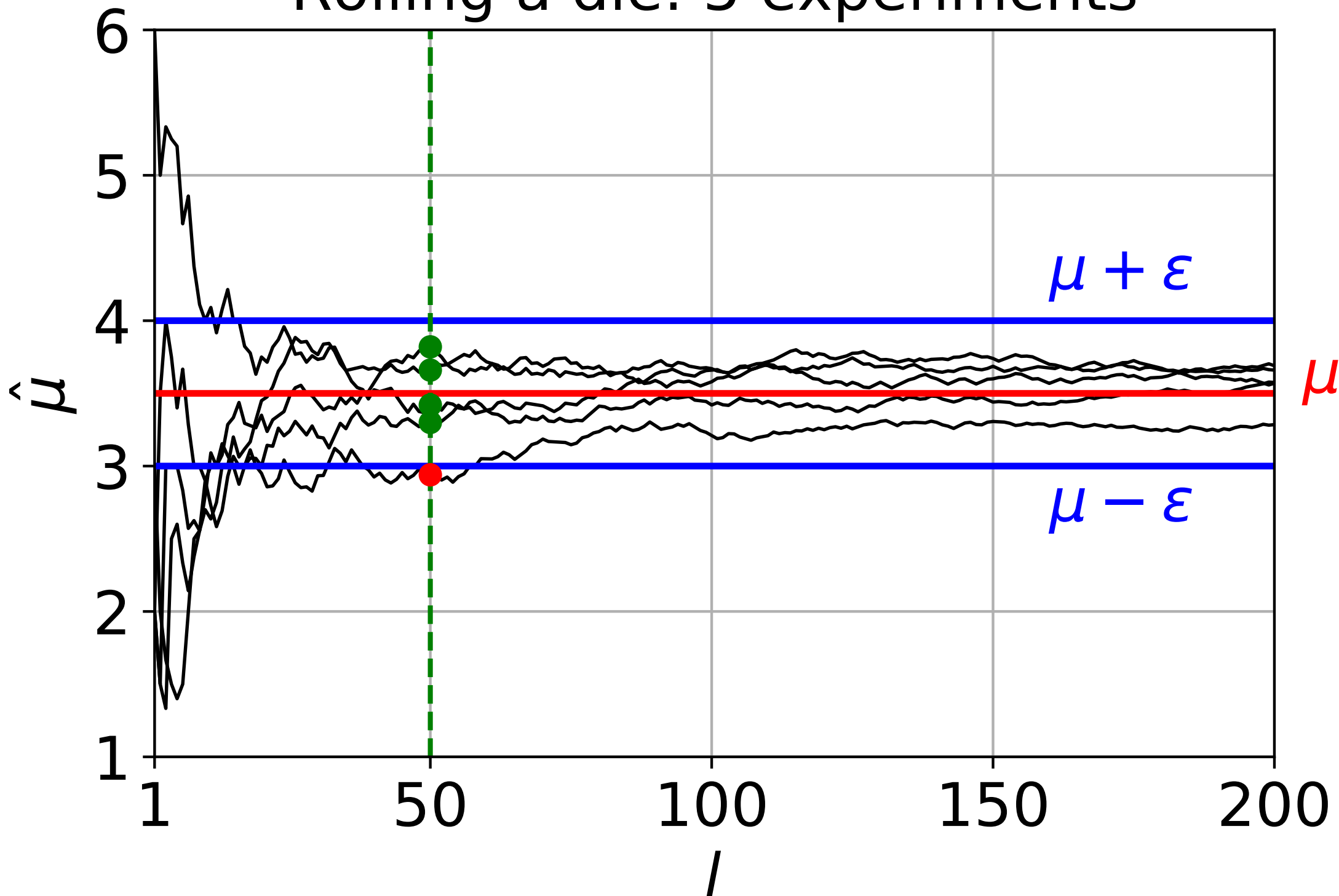
Rolling a die: 5 experiments



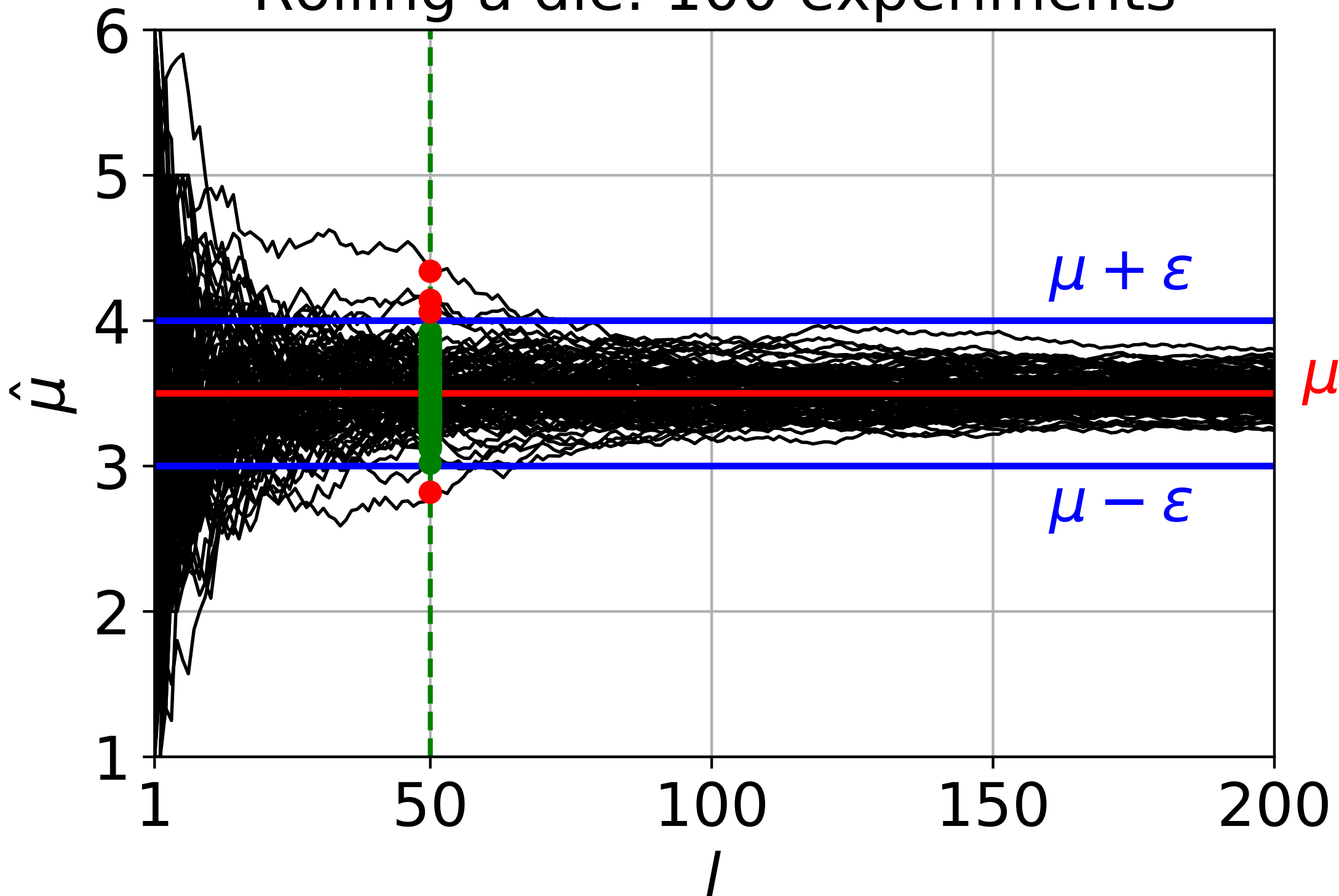
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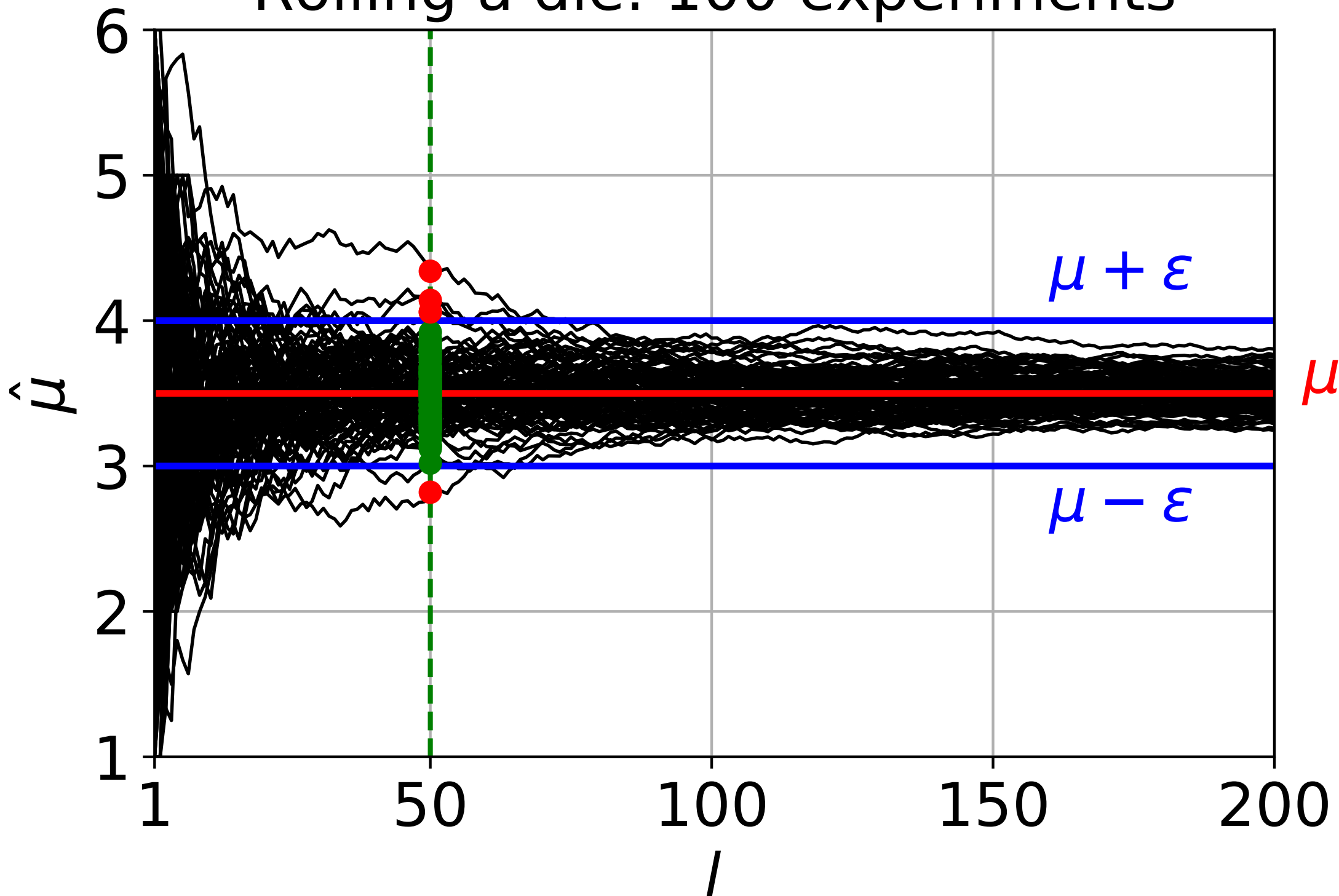
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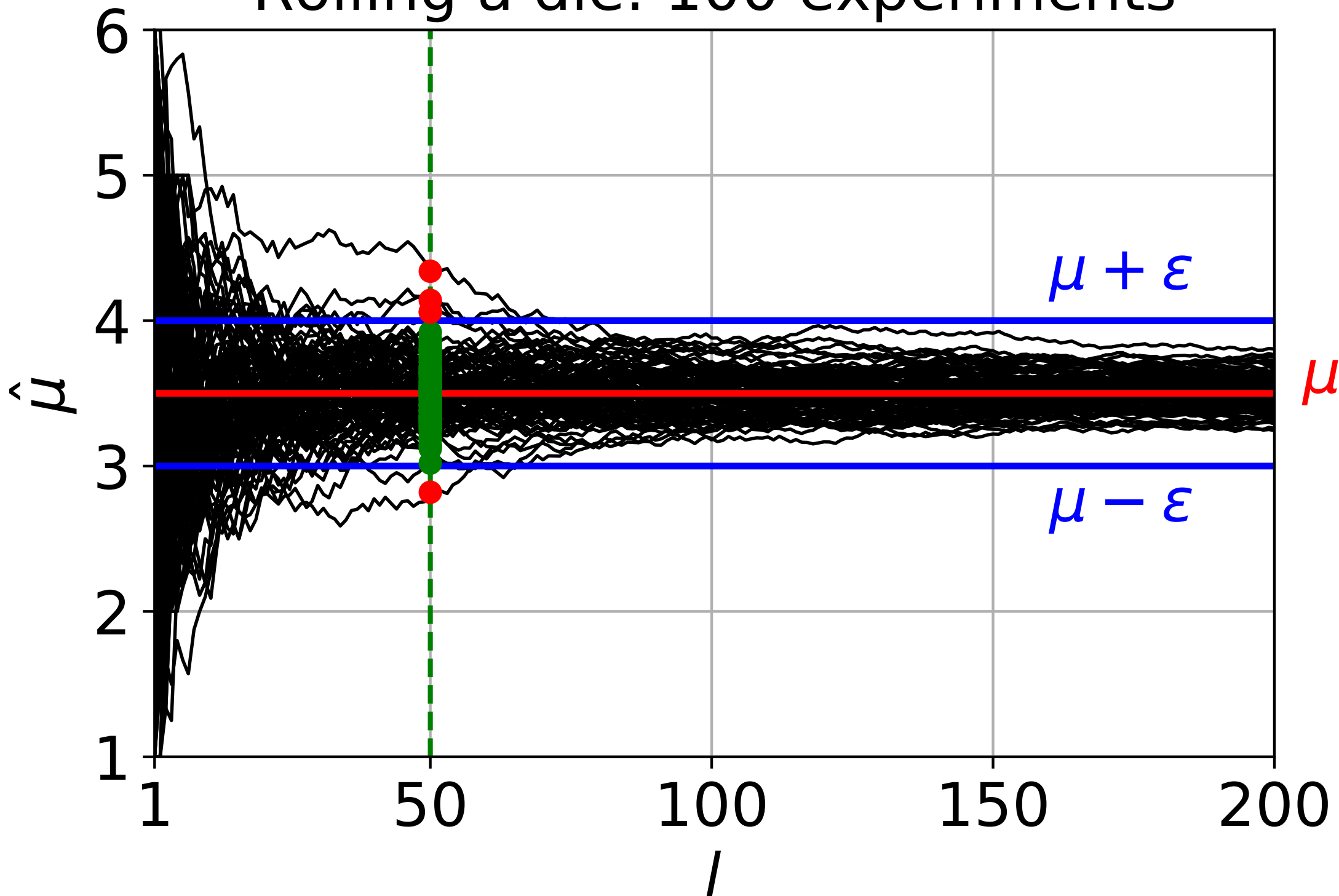
Rolling a die: 100 experiments



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